

UPSC Prelims

Science & Tech

Class No. 14

Emerging Technology

Non Fungible Tokens

- Digital identifier, similar to a certificate of ownership that represents a digital or physical asset; financial security consisting of digital data stored in a blockchain, a form of distributed ledger;
- NFT market grew exponentially from 2020–2021: the trading of NFTs in 2021 increased to more than \$17 billion, up by 21,000% over 2020's total of \$82 million; Work entitled Merge by artist Pak was one the most expensive NFT, with an auction price of US\$91.8 million.
- What is fungibility?

Fungibility

- Fungibility means subdivision into units as well as mutual substitution
- Example : Rs. 2000 note.
- Subdivided : yes, 500 x 4 notes
- Mutual substitution : yes, both have the same value
- Other cases :
- 1 kg gold bar
- 100 gm diamond
- NFT : not possible

NFT

- NFT is a Digital file such as photo file(JPEG),Animated image(GIF),music file (MP3) etc. Stored using blockchain Technology.
- They cannot be sub divided and/or their individual sub-units can not be exchanged with one another. Because their values are different based on buyer's preference. So NFTs are non- fungible.
- How does it work?

NFT

- Advantages :
- Multimedia creators : trading of digital assets
- Easier to verify opwnership records
- Paper less administrations : digital torage of tickets / records
- Challenges :
- mass hysteria of investment bubble might collapse leading to bankruptcy of gullible citizens
- Money laundering and terror finance : same as bitcoin
- Electricity consumption and environmental challenges

Internet Of Things

- It is a **seamless connected network of embedded objects/ devices, with identifiers, in which Machine to Machine (M2M) communication without any human intervention is possible** using standard and interoperable communication protocols.
- In general terms, it includes **any object or thing that can be connected to an Internet network**, from factory equipment and cars to mobile devices and smart watches.
- **Phones, Tablets and PCs** are not included as part of IoT.

Internet Of Things

- It specifically comes to mean **connected things that are equipped with sensors, software, and other technologies** that allow them to transmit and receive data.
- It is widely being used to create **smart infrastructure in various verticals such as Power, Automotive, Safety and Surveillance, Remote Health Management, Agriculture, Smart Homes and Smart Cities** etc, using connected devices.

How does it work?

- An IoT ecosystem consists of **web-enabled smart devices use embedded systems, such as processors, sensors and communication hardware**, to collect, send and act on data they acquire from their environments.
- These **share the sensor data they collect by connecting to an IoT gateway or other edge device** where data is sent to the cloud to be analyzed locally.

How does it work?

- Sometimes, **these devices communicate with other related devices** and act on the information they get from one another.
- The devices do most of the work without **human intervention**, **although people can interact with the devices for instance, to set them up, give them instructions or access the data.**

Internet of Things Uses



Significance and challenges

- IoT encourages the communication between devices, also known as **M2M communication**.
- M2M communication helps to **maintain transparency** in the processes. It leads to **uniformity in the tasks** and **maintain the quality of service**.
- M2M interaction provides **better efficiency**, hence, **accurate results can be obtained fast**.
- Applications of IoT in increased **convenience**, **better management**, thereby **improving the quality of life**.

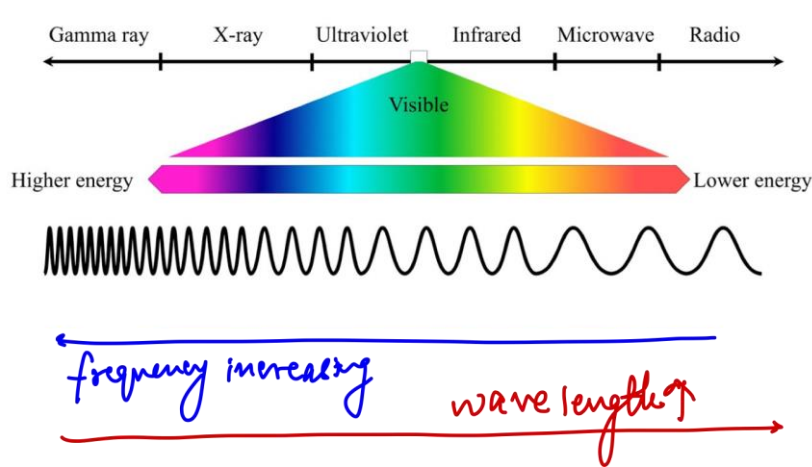
Significance and challenges

- **Challenges of IoT:** Data Breach, increased dependence on Technology, Complexity in operation, issue of compatibility in tagging and monitoring, lesser Employment of Menial Staff etc.
- Further challenges : internet consumption and bandwidth issues
- Solutions : Lifi , 5g

Electromagnetic Spectrum Explained

Electromagnetic Spectrum

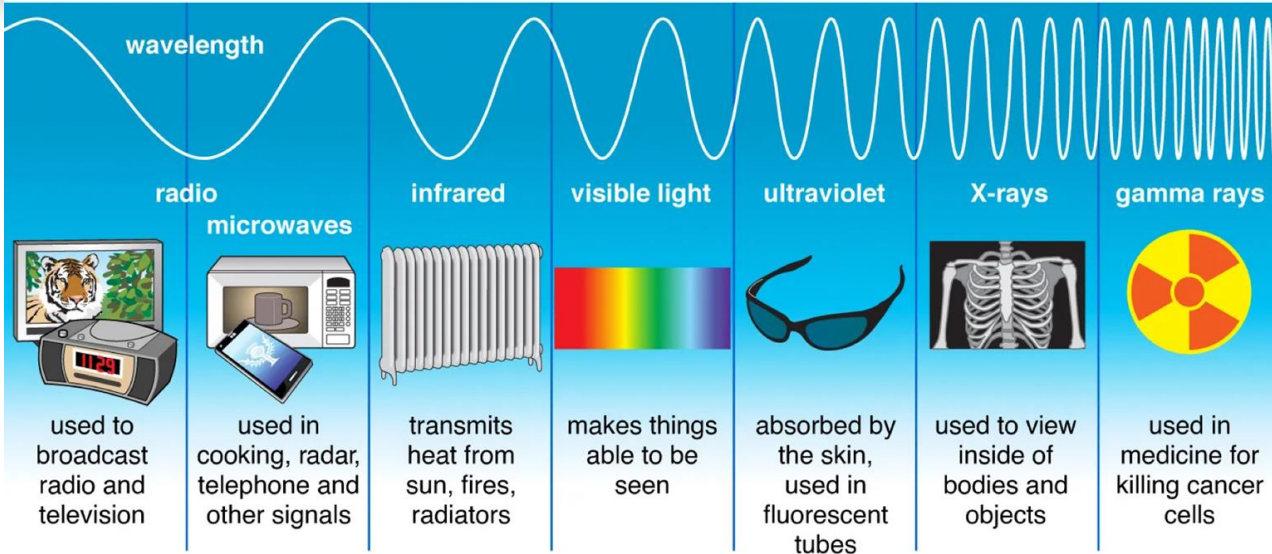
Full 'spectrum' of radiation emitted from a star.



$$E = h \frac{c}{\lambda} = h\nu$$

Electromagnetic Spectrum Explained

Types of Electromagnetic Radiation



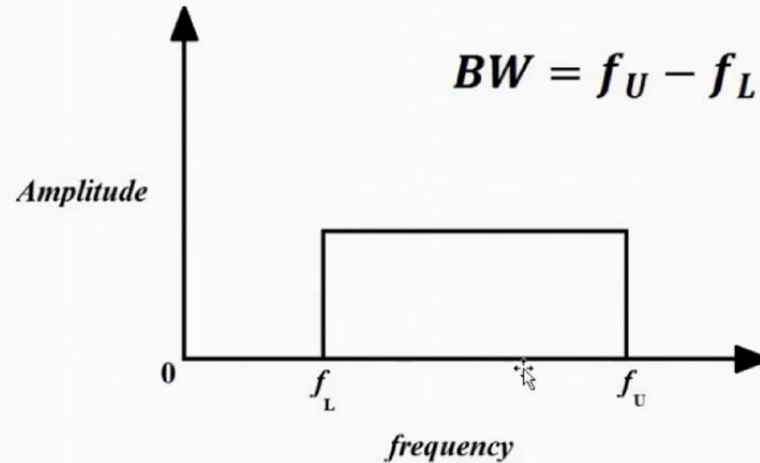
Electromagnetic Spectrum Explained

<u>Region</u>	<u>Frequency</u>	<u>Wavelength</u>
Radio	$< 3 \times 10^9 \text{ Hz}$	$> 10 \text{ cm}$
Microwave	$3 \times 10^9 - 3 \times 10^{11} \text{ Hz}$	$10 - 0.1 \text{ cm}$
Infrared	$3 \times 10^{11} - 4 \times 10^{14} \text{ Hz}$	$1000 - 0.7 \mu\text{m}$
Visible	$4 \times 10^{14} - 7.5 \times 10^{14} \text{ Hz}$	$700 - 400 \text{ nm}$
Ultraviolet	$7.5 \times 10^{14} - 3 \times 10^{16} \text{ Hz}$	$400 - 10 \text{ nm}$
X-ray	$3 \times 10^{16} - 3 \times 10^{19} \text{ Hz}$	$10 - 0.01 \text{ nm}$
γ -ray	$> 3 \times 10^{19} \text{ Hz}$	$< 0.01 \text{ nm}$

Electromagnetic Spectrum Explained

What is Bandwidth ?

- *Bandwidth* is the *portion of the electromagnetic spectrum occupied by a signal.*
- It is the *frequency range over which signal is transmitted.*



LiFi technology

- LiFi is a wireless optical networking technology, which **uses light emitting diodes (LEDs) to transmit data.**
- It makes a LED light bulb emit pulses of light that **are undetectable to human eye and within those emitted pulses, data can travel** to and from receivers.

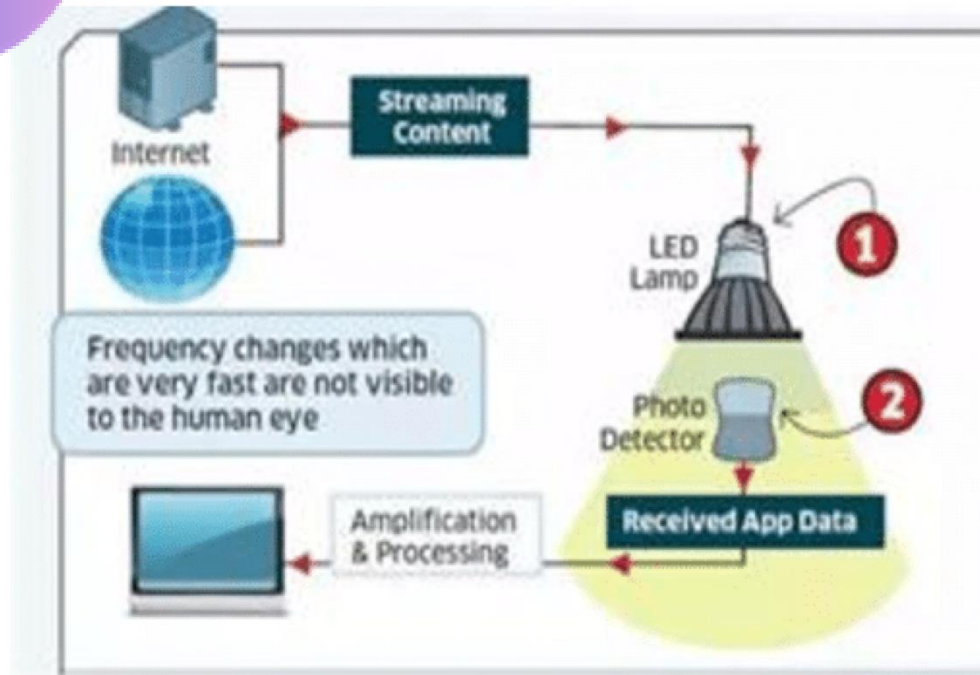
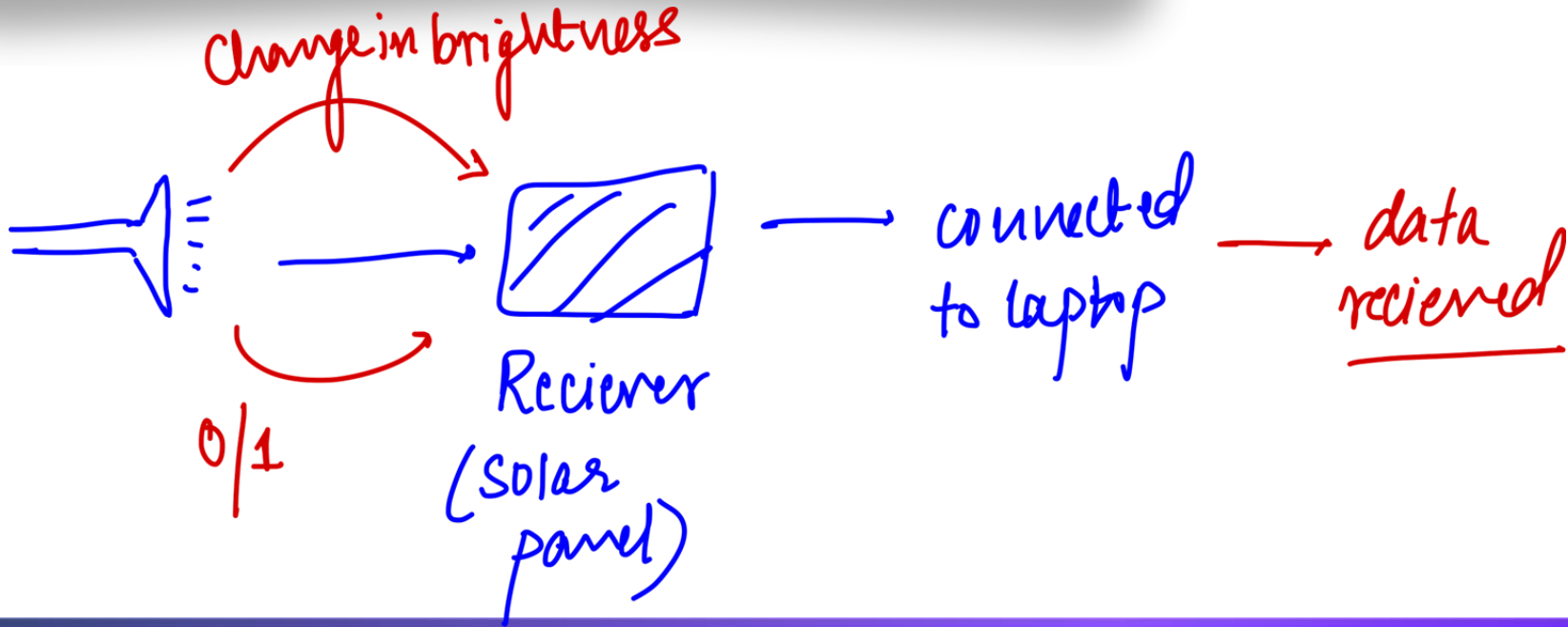


Figure 2: Procedure of Li-fi Technology

LIFI explained



LIFI technology

- Not a strictly line of sight technology : data rate is not dependent on line of sight but on signal quality of device.
- **Wifi : uses radio waves, Lifi : uses light.**

LIFI technology

Visible Light Communication (VLC) technology



VLC uses **electromagnetic spectrum wavelengths 375 to 780 nm.**



It is possible to **achieve illumination and data transfer simultaneously** by means of LEDs that is the prominent lighting equipment lately.



VLC **can transmit large amounts of data** faster than Bluetooth.



VLC has no **electromagnetic interference.**

Feature	LiFi	WiFi
Operation	LiFi transmits data using light with the help of LED bulbs.	WiFi transmits data using radio waves with the help of WiFi router.
Interference	Do not have any interference issues similar to radio frequency waves.	Will have interference issues from nearby access points (routers).
Privacy	In LiFi, light does not pass thorough the walls and hence will provide a much secure data transfer	In WiFi, Radio frequencies signal passess thorough the walls and hence there is a need to employ techniques to achieve secure data transfer.
Data transfer speed	About 1 Gbps	About 150Mbps.
Frequency of operation	10 thousand times frequency spectrum of the radio	2.4GHz, 4.9GHz and 5GHz
Coverage distance	About 10 meters	About 32 meters (vary based on transmit power and antenna type)
Data density	work with high dense environment	work in less dense environment due to interference related issues

5G technology

- 5G (the fifth generation of cellular networks) is designed to improve network connections by **addressing the legacy issues of speed, latency and utility**, which the earlier/current generation of mobile networks could not address.
- **5G operates at higher frequencies** to offer a new kind of network that is **designed to connect virtually everyone and everything together** including machines, objects, and devices.

5G technology

- It will also have an **enhanced throughput to handle more simultaneous connections** at a time than current-generation networks.
- 5G **mainly works in 3 bands**, namely low, mid and high frequency spectrum.

3G vs 4G vs 5G vs 6G

				
	Deployment	Bandwidth	Latency	Speed Average
3G	2004-06	2mbps	100-150 millisecond	144 kbps
4G	2006-10	200mbps	20-30 millisecond	25 mbps
5G	2020	>1 gbps	<10 millisecond	200-400 mbps
6G	2028-2030	1 tbps	<1 microsecond	About 50 times faster than 5G

5G is the Fifth generation of cellular networks.

5G Spectrum Types

5G Spectrum	Features
 Low-band	▶ Lower than 1 GHz on the spectrum chart. ▶ Wider coverage but limited improvement in speed and latency. ▶ For thinly populated areas and provides indoor coverage in built-up areas.
 Mid-band	▶ 1 GHz - 6 GHz range. ▶ Ideal for 5G because it can carry plenty of data while also traveling significant distances.
 High-band	▶ 24 GHz band and higher. ▶ It delivers super-fast speeds over short distances.

5G technology

Spectrum	Uses/Advantage	Limitations
Low Band	• Great promise in terms of coverage and speed of internet and data exchange, the maximum speed is limited to 100 Mbps (Megabits per second) • For commercial cellphone users who may not have specific demands for every high speed internet.	• May not be optimal for specialised needs of the industry.
Mid-Band	• High speeds compared to the low band. • May be used by industries and specialised factory building captive networks.	In terms of coverage area and penetration of signals.
High-band	• Offers the highest speed (as high as 20 gbps) (giga bits per seconds) of all the three bans.	• Extremely limited coverage and signal penetration strength.

5G technology

Remember : What is the problem of having a high frequency communication band?

5G technology : challenges in India

Challenges to 5G evolution in India



Lack of infrastructure

(inadequate optical fiber infrastructure and backhaul, interrupted power supply ,etc)



High Base Price for Auction



Telco issues

(Low Average revenue per user, Lack of skillset etc.)



Consumer constraints

(Network coverage issue, handset availability, data privacy etc)



Technological Challenges

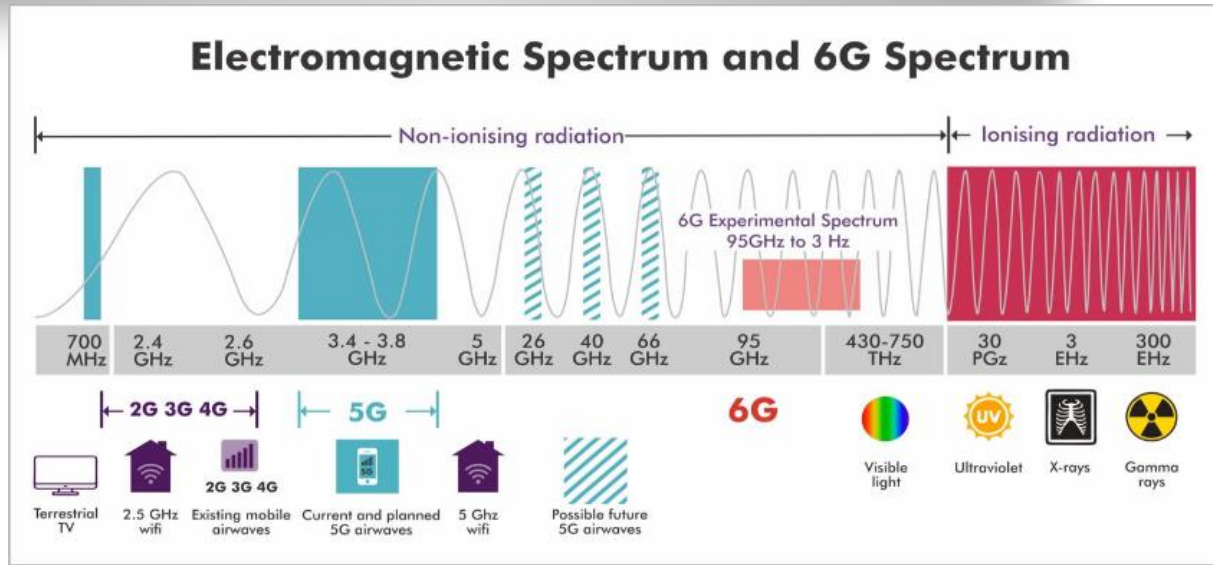
(High import dependency on complete 5G Supply Chain etc.)



Other

(Connectivity with critical infrastructure, National security issue from chinese vendors like ZTE, Huawei etc)

5G and EM spectrum



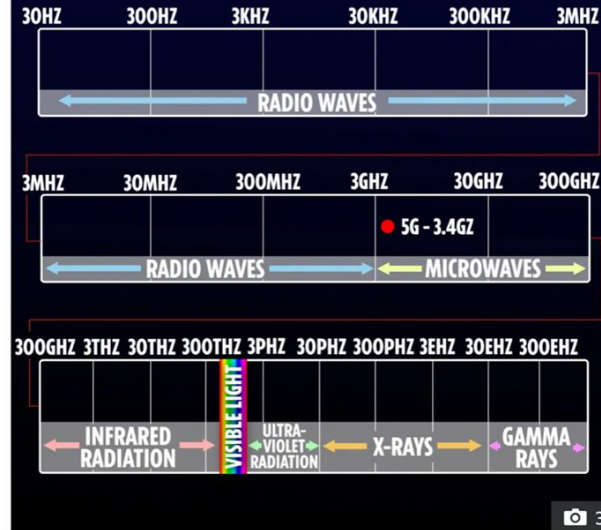
5G and EM spectrum

^ Electromagnetic spectrum

Microwaves occupy a place in the electromagnetic spectrum with frequency above ordinary radio waves, and below infrared light:

Electromagnetic spectrum			
Name	Wavelength	Frequency (Hz)	Photon energy (eV)
Gamma ray	< 0.01 nm	> 30 EHz	> 124 keV
X-ray	0.01 nm – 10 nm	30 EHz – 30 PHz	124 keV – 124 eV
Ultraviolet	10 nm – 400 nm	30 PHz – 750 THz	124 eV – 3 eV
Visible light	400 nm – 750 nm	750 THz – 400 THz	3 eV – 1.7 eV
Infrared	750 nm – 1 mm	400 THz – 300 GHz	1.7 eV – 1.24 meV
Microwave	1 mm – 1 m	300 GHz – 300 MHz	1.24 meV – 1.24 μeV
Radio	≥ 1 m	≤ 300 MHz	≤ 1.24 μeV

ELECTROMAGNETIC SPECTRUM



5G is a very low-frequency form of radiation – far below visible light and infrared

5G in news

- Hon'ble PM has inaugurated country's first 5G testbed.
- Multi- Institute collaborative project lead by IIT madras.
- 5G has an amalgamation of various technologies :
- Massive Multi User MIMO (multiple input multiple output) enabled network.
- Small Cell Stations : connect base stations and users.
- Mobile Edge computing.
- Beamforming

5G : related terms

- Network Slicing : multiple networks to be created on top of a common physical infrastructure. Each 'slice' can be allocated based on the specific needs of the application, use case or customer.
- Fiberisation : Connecting radio towers with each other via optical fiber cables. Connects core of network to edge facilitating large amounts of data used in 5G services.
- 5G airwave interference : 5G uses the same spectrum as aircraft radio altimeters leading to interference.

5G : spectrum auction

- Auction of 5G spectrum was held recently worth ₹1.5L crore.
- It relates to the radio frequencies allocated to the mobile industry and other sectors for communication over the airwaves.
- Spectrum is a sovereign asset.
- Use of mix of low band, mid band and high band spectrum to deliver tailored 5G experience.

5G : open radio access network

- Centre for development of telematics along with 2 private networks has signed agreement for developing 5G open RAN.
- RAN : part of network that connects main network infrastructure to end users.
- Connects individual devices to other parts of a network through radio connections. It acts as a bridge to access all key applications on the web.
- Current RAN is integrated platform of both hardware and software, therefore it is difficult to mix vendors.
- Open RAN will allow vendors to mix and match components from different vendors, leading to a multi supplier RAN solution.

Long term Evolution (LTE) and Voice over LTE (VoLTE)

- VoLTE is a **technology update to the LTE protocol** used by mobile phone networks.
- **Under LTE**, the infrastructure of telecom players **only allows transmission of data while voice calls are routed to their older 2G or 3G networks.**
- This is why, under LTE, you cannot access your 4G data services while on a call.

Long term Evolution (LTE) and Voice over LTE (VoLTE)

- This leads to problems Such as slow internet speeds and poor voice clarity.
- LTE is commonly marketed as 4G LTE.
- **VoLTE allows voice calls to be ‘packaged’ and carried through LTE networks.** This would mean 4G data accessibility even during calls.

Metaverse

- Metaverse projects **refer to blockchain-based games and applications** that are set in virtual worlds.
- Such games follow a **play-to-earn model, where gamers are rewarded for being invested** in the game.
- Metaverse can be understood as a combination of multiple elements of technology, including virtual reality, augmented reality and video where users "live" within a digitally enhanced surrounding.

Metaverse

- **These technologies** are currently being developed by Apple, Google, Amazon, and Microsoft in addition to Facebook.
- Metaverse uses the following technologies : (extended realities)
- Virtual Reality
- Augmented Reality
- Hologram
- Avatar
- Let's see a video on how it works

Metaverse : Technologies

- **Virtual Reality (VR):** VR can be understood as the use of computer modelling and simulation that enables a person to interact with an artificial three-dimensional (3-D) visual or other sensory environment. E.g., games like World of Warcraft. **VR closes the world**, and transposes an individual, providing complete immersion experience.

Metaverse : Technologies

- **Augmented Reality (AR):** It is an enhanced version of the real physical world that is achieved through the use of digital visual elements, sound, or other sensory stimuli delivered via technology. E.g., games like Pokémon Go.
- **Hologram:** Holograms are virtual three-dimensional images created by the interference of light beams that reflect real physical objects.

Metaverse : Technologies

- **Avatar:** An avatar in the metaverse is a representation of an individual in the virtual world, this digital avatar enables the person to function like an actual human being in a digitally created world.
- **Platform/Developer/User distinction:** In metaverse, digital engagement will get very personal and tailormade for the user. In this context, it is important that role played by the platform, the developers and the users is clearly understood.

Metaverse : Technologies : Images

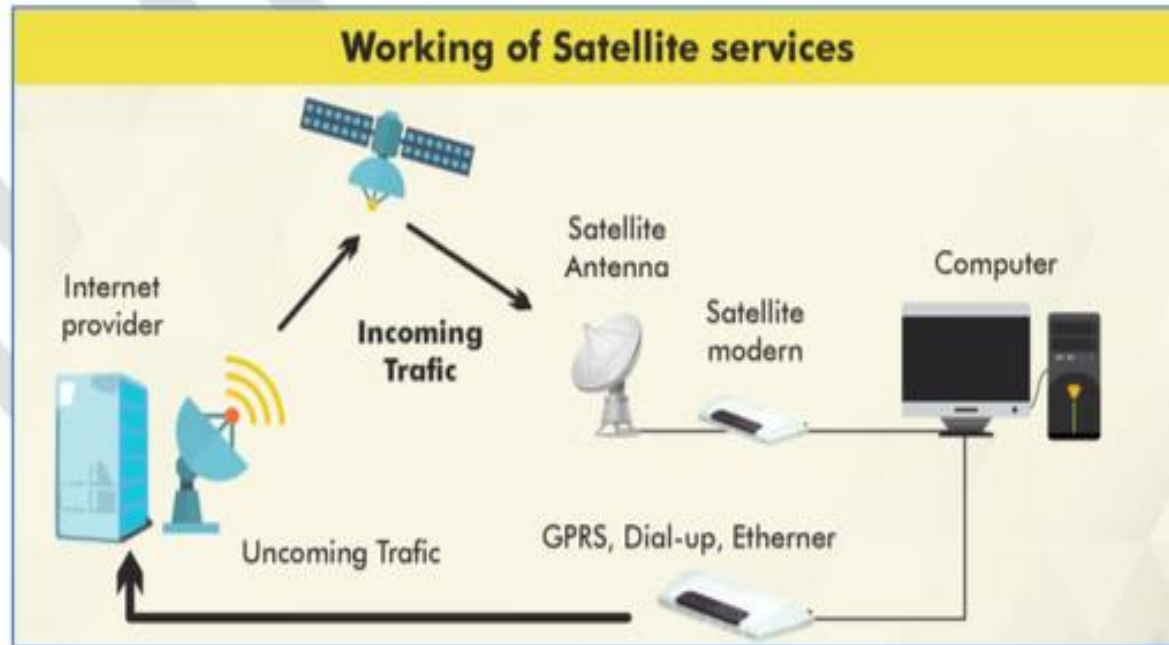


Satellite based Internet Connectivity

- Satellite Internet Services is a **wireless internet connection** which uses satellites in space to get an internet signal from the Internet Service Provider (ISP) to users.

Satellite based Internet Connectivity

- It operates using two way satellites like **VSAT (Very Small Aperture Terminal)** or telecommunication satellites to liaison between Internet Service Providers (ISPs) and Users with satellite dishes.
- **Direct to Satellite Model:** allows devices to directly communicate with the satellite without the need of any intermediate ground gateway.
- 2 types : satellite in LEO or satellite in GEO



Satellite based Internet Connectivity

- **Advantages of Space-based internet systems using satellites in LEO compared to GEO:** low Latency in signal transmission.
- **Advantages of Space-based internet systems using satellites in GEO compared to LEO:** Serviceable area on Earth by each satellite is more, easier to establish a transmission link with a satellite.

Models for Satellite based connectivity

- **Hybrid : LPWAN + Satellite** : each sensor first connects with a sink node, (LPWAN) then to Satellite.
- **Direct to Satellite** : Devices to directly communicate with satellite without need of ground gateway.



Satellite-based connectivity is **possible using Geo Stationary, Medium and Low Earth orbit Satellites**



It is **different from satellite television** as Internet signals go **both ways** with a much **greater amount of bandwidth** to move all that data



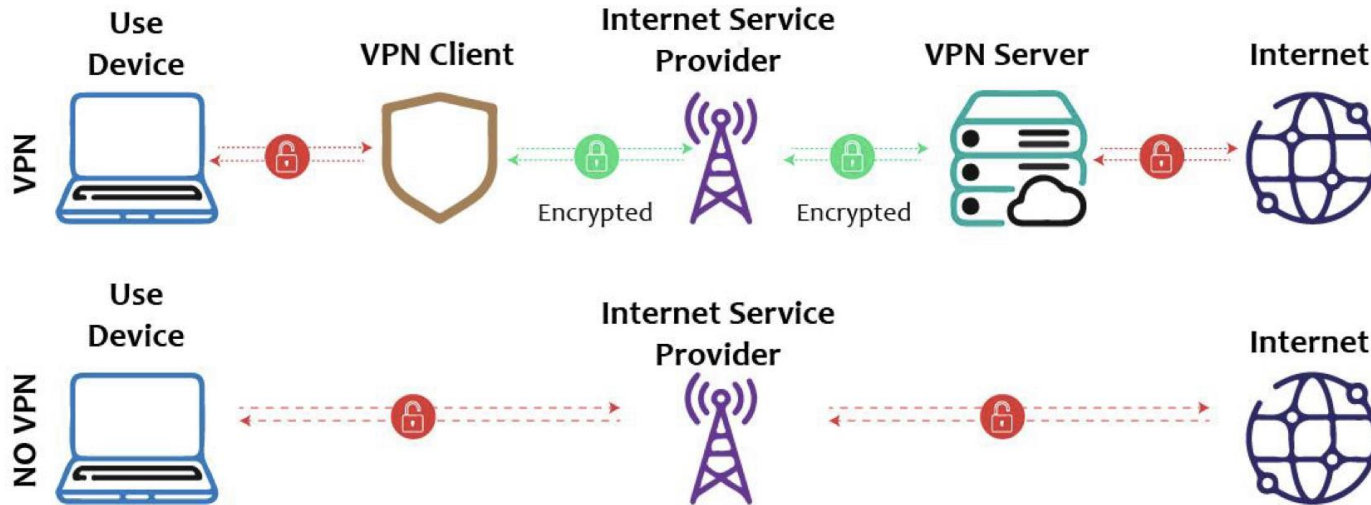
Broadband **signal is sent and received to a satellite in space** rather than using an **underground copper/fibre network**

Satellite based Internet Connectivity

Benefits of Satellite internet Services	Challenges in the Satellite internet Services
• Cheap and easy internet option for rural area, hilly terrain, and islands for last mile connectivity. • Faster than DSL connection with continuously improving technology with present speeds reaching around 100 mbps. • Can play a significant role in emergency or disaster recovery due to limited ground infrastructure.	• Low speed and high latency - In comparison to fiber optic cables, • High cost as compared to wireless cellular network and others. • Limitations of Bandwidth with high impact of weather aberrations • Potential to damage spacecrafts/satellites or interrupt of higher satellites frequency because of its large satellite network need, • Potential of increase space junk. • A VPN gives online privacy anonymity by creating a private network from a public internet connection

Virtual Private Networks

- A VPN is a computer network across a public internet that provides user access to their organization's network while maintaining the security of information provided
- It gives online privacy and anonymity by creating a private network from a public internet connection.





Types of VPN



Secure Sockets Layer (SSL) VPN

It enables individual users to **access an organization's network** and client-server applications **without the need for specialized software**



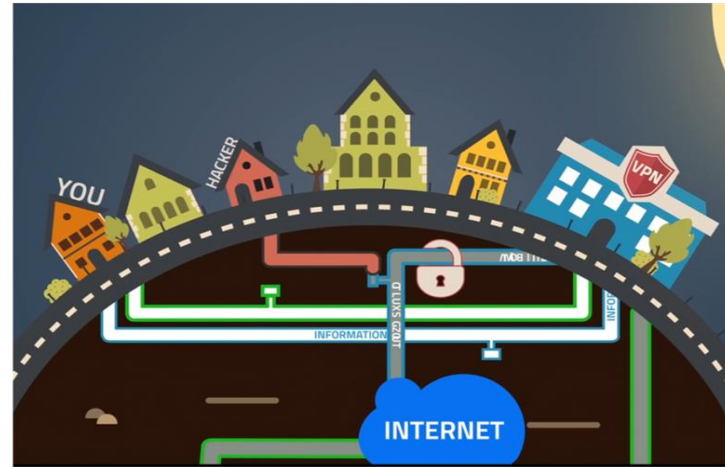
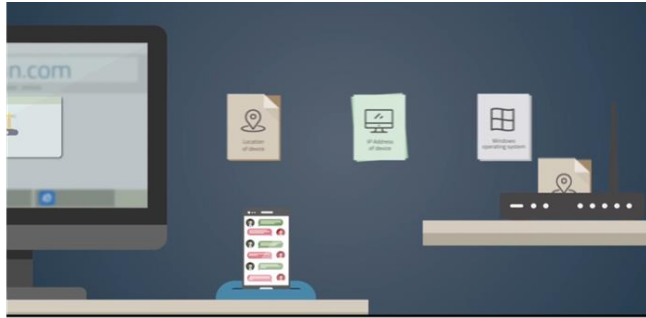
Site-to-site VPN

A **private network designed to hide private intranets** and allow users of these secure networks to access each other's resources.



Client-to-Server VPN

User is not connected to the **internet via his own ISP** (internet service provider), but establishing a **direct connection through VPN provider**.



Dark Net

- Imagine you get an email containing your bank statement :
- It is on the internet
- It is not accessible to me.
- This is the deep web, it is that part of Internet which cannot be accessed through traditional search engines like Google nor is it accessible by normal browsers like Chrome or Safari.
- Dark Net is a part of deep web

Dark Net

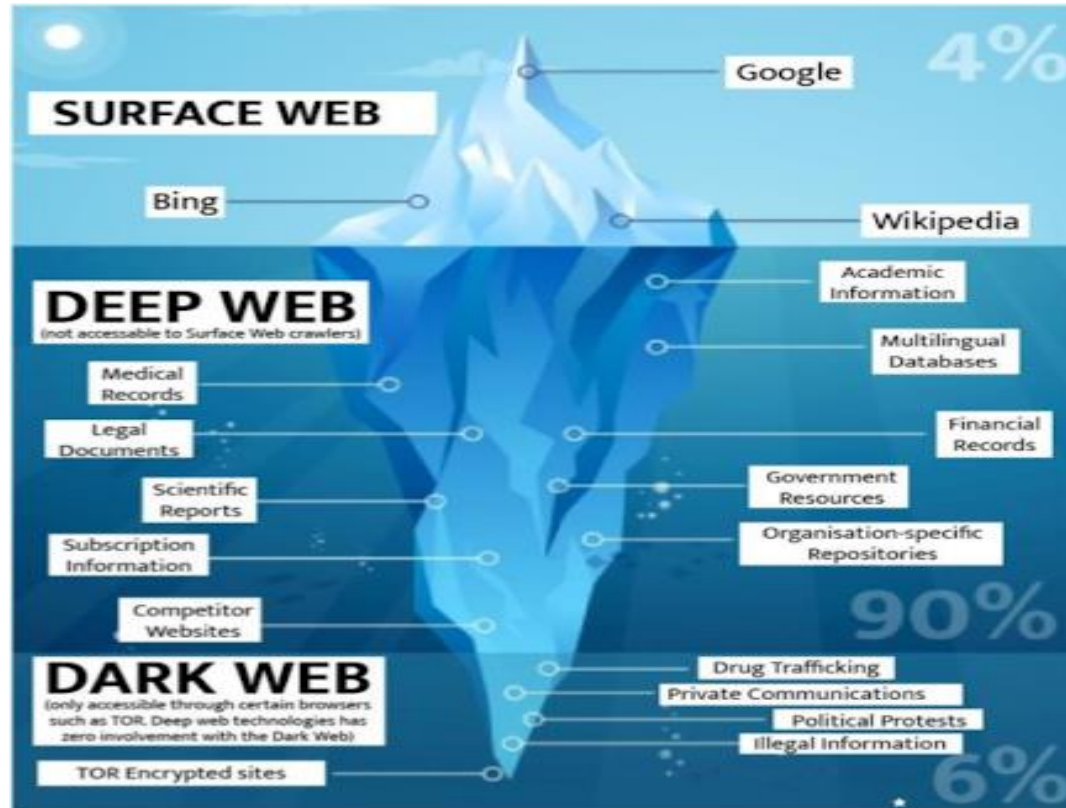
- It generally uses non-standard communication protocols which make it inaccessible to internet service providers (ISPs) or government authorities.
- Content on Dark Net is encrypted and requires specific browser such as TOR (The Onion Router) browser to access those pages.

Dark Net

- Dark Net itself is only a part of Deep Web that is a broader concept, which includes sites that are protected by passwords.
- Part of internet that is readily available to general public and searchable on standard search engines is called as Surface Web.

Dark Net

- Concerns over its use: Anonymity, Haven for illicit activity, Privacy and ethical concerns, drug dealing, communication by terrorists etc.

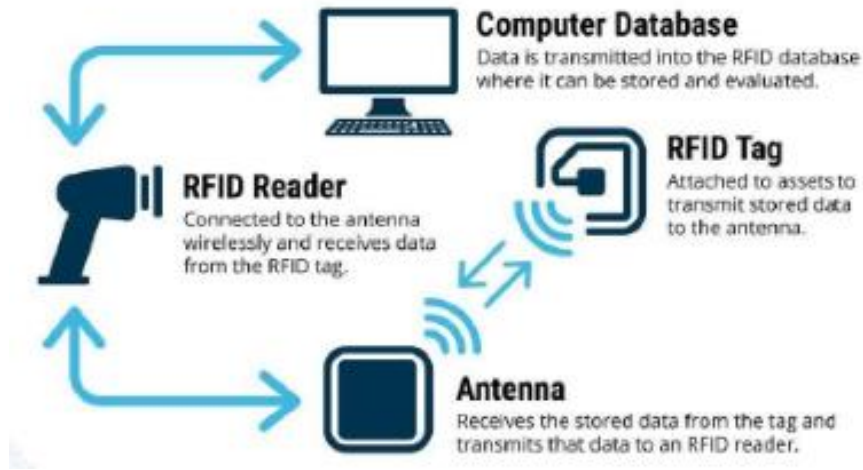


RFID

- It is an electronic tag that can exchange data with an RFID reader through radio waves.
- In this, an antenna broadcasts energy to tag, which in turn returns modulated energy back in form of back-scatter.
- It is faster connection, and operational range is about 100m.
- RFID tag typically sends out information hence only one-way information.

RFID

Basic RFID System

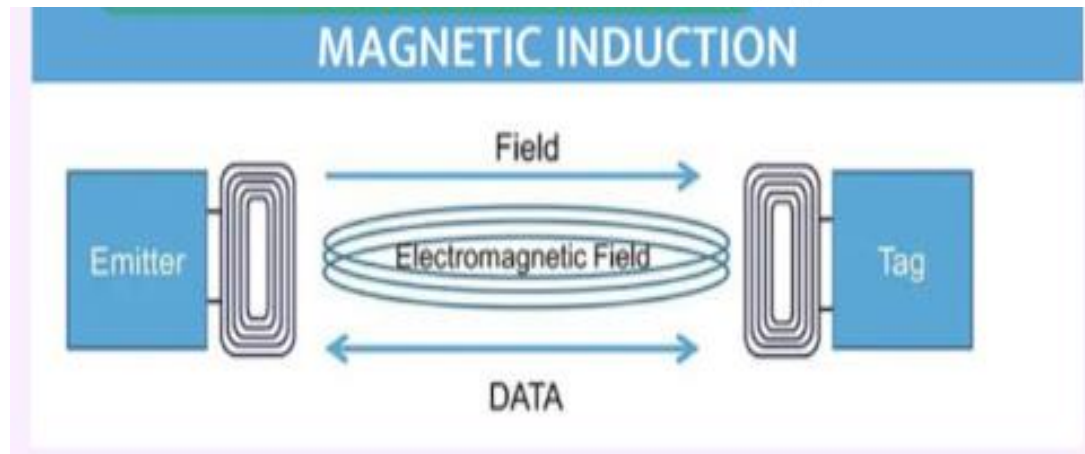


Near Field Communication

- NFC is the technology that allows compatible devices to establish radio communication with each other by touching them together or bringing them into a distance maximum of 10 cm.
- NFC works on Radio Frequency Identification (RFID) technology and Magnetic induction principle
- However, RFID and NFC also have many differences

Near Field Communication

- When electrons flow through a conductor, they create a magnetic field. And when magnetic fields change, they can cause electrons to flow through a conductor. This relationship -- known as inductive coupling.
- In NFC also the reader or the mobile acts as the active device whose antenna or the coil releases a sinusoidal carrier wave and creates a magnetic field. Then the passive tags present in that field gets energy from the reader and change the properties of the receiving frequency.



Basis	RFID	NFC
Communication	Unidirectional—typically an RFID tag sending data to an RFID reader	Both unidirectional and bidirectional. This allows NFC to be used for more complex interactions such as peer-to-peer (P2P) data sharing as well as card emulation.
Frequency range	RFID FREQUENCY RANGES: Low Frequency (LF): 125-134 KHz (LF) Range: Up to 10 m High Frequency (HF): 13.56 MHz NFC (HF) Range: Up to 30 m Ultra High Frequency (UHF): 855 MHz TO 960 MHz (UHF) Range: Up to 100 m	
Application	Asset tracking Race timing Attendee tracking Inventory management	Contactless payment Information/data sharing In-store check-ins wearables like smartwatch Post-sale product experiences (such as scanning QR codes on posters for additional product information)

	Radiofrequency Identification (RFID) 	Quick Response (QR) code 	Near Field Communication Technology (NFCT) 
About	A wireless system comprised of two components: tags and readers.	A two-dimensional version of a barcode that can be read easily by a digital device	A short-range wireless connectivity technology that allows NFC-enabled devices to communicate with each other and transfer information quickly and easily with a single touch
Working Range	Read distance depends upon the frequency (Ranging from 300 gigahertz to as low as 9 kilohertz) used	Ideal scanning size to distance ratio is 1:10.	NFC-enabled devices must be either physically touching or within a few centimeters from each other for data transfer.
Other Key Information	Do not require a direct line of sight to be read. - Data stored in an RFID tag can be updated in real-time.	Require a line of sight to scan. No real-time tracking.	Transfer information quickly and easily with a single touch.

Other Similar Technology

- Infrared : short connection and transmission of information through infrared light.
- Needs line of sight between 2 devices.
- Ex : Remote control devices.

Edge Computing

- It enables data to be analysed, processed, transferred at edge of a network.
- It means, data is analysed locally, closer to where it is stored, in real-time without latency.
- Basic difference between edge computing and cloud computing lies in where data processing takes place

Edge Computing

- Cloud Computing: It means storing and accessing data and programs over Internet instead of your computer's hard drive.
- Existing Internet of Things (IoT) systems perform all their computations in cloud using data centres.
- Edge Computing: It manages massive amounts of data generated by IoT devices by storing and processing data locally.

Edge Computing

Benefits :

- High speed
- Reduced latency
- Better reliability and security : difficult for any single disruption to take down a network
- Very important in upcoming internet of things technology

4th IR

- 1st IR : ?
- 2nd IR ?
- 3rd IR ?
- 4th IR :
- Building on the 3rd Industrial Revolution
- Includes cyber-physical systems, the Internet of things, big data analytics, cloud computing, cognitive computing, artificial intelligence, 3-D printing, and autonomous vehicles etc.
- Impact : all of the above

Big Data

- Big data is a term applied to datasets whose size or type is beyond the ability of traditional relational databases to capture, manage and process the data with low latency.
- Big data analytics is the use of advanced analytic techniques against very large, diverse data sets that include structured, semi-structured and unstructured data, from different sources, and in different sizes from terabytes to zettabytes.



Big Data : Advantages

- Improves the decision making of businesses
- Protects company and client information
- Effective marketing and targeted advertising
- Cost and time reduction through streamlining operations
- Better Products designing by getting info and analysis
- Applications in various sectors :
 - Banking : NPA prevention and trend analysis
 - Education : identify at risk students etc.
 - Government : traffic congestion and utilities management
 - Health care : insights to improve patient care
 - Manufacturing : boost quality output while minimizing waste processes
 - Agriculture : Optimizing crop efficiency

Big Data

- Challenges :
- Human resource : lack of data scientists
- Privacy Issues : reveal sensitive and personal information of users
- Security issues and outsourcing of data analytics
- Lack of availability of quality data
- Ethics and issues of informed consent of users

Additive Manufacturing (3d printing)

- Three-dimensional printing (3D printing), also known as Additive manufacturing (AM) is a process of joining materials to make objects from 3D model data, usually layer upon layer.
- Different from conventional manufacturing
- It is a process in which a three-dimensional object is built from a computer-aided design (CAD) model, usually by successively adding materials in a layer-by-layer fashion.
- The addition of material can happen in multiple ways, namely power deposition, resin curing, filament fusing.

Additive Manufacturing (3d printing)

- Thermoplastics, Metals, Ceramics as well as biomaterials can be used for additive manufacturing.
- Traditional vs. AM: Unlike AM, traditional manufacturing methods, by contrast, are subtractive in nature.
- Subtractive manufacturing involves removing parts of a block of material in order to create the desired shape. Cutting wood into useful shapes, for instance, is a very simple example of a subtractive process.

3D printing

Basic Additive Manufacturing Process

CAD Model.....3D Object



3D Card Model



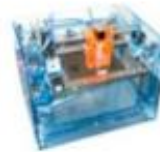
.STL File



Slicing Software



Layer Silces & Tool Path



AM Process



3D Object

Benefits

- Low cost
- Less time
- Efficiency
- Customization
- Increased Productivity
- Reduced Wastage

Issues

- Size is limited
- Limited raw materials
- Employment destruction
- Copyright infringement concerns
- Dangerous items can be produced

Artificial Intelligence

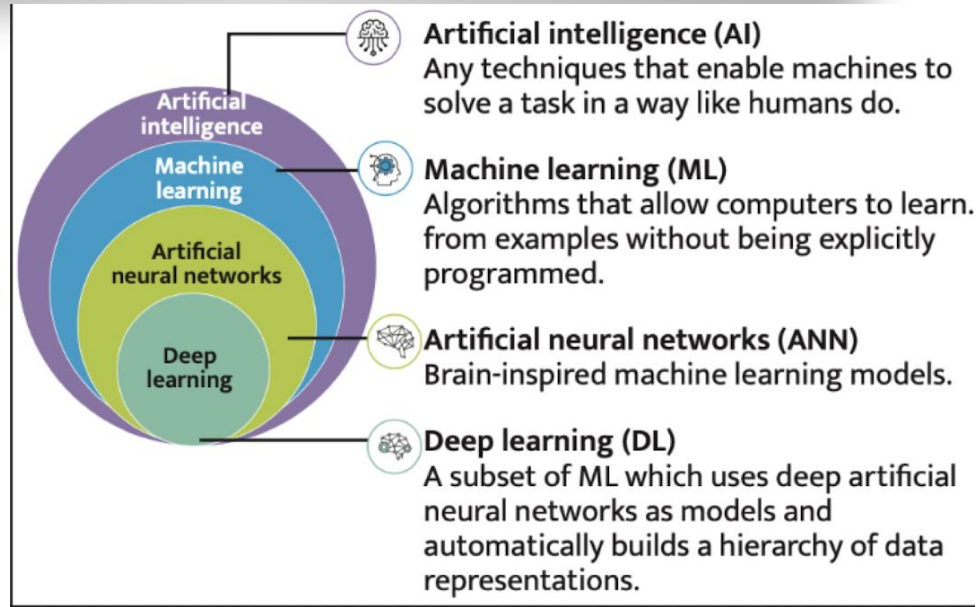
Simulation of Human Intelligence processes by machines

- Cognitive skills
 - Learning
 - Reasoning
 - Self correction

Artificial Intelligence

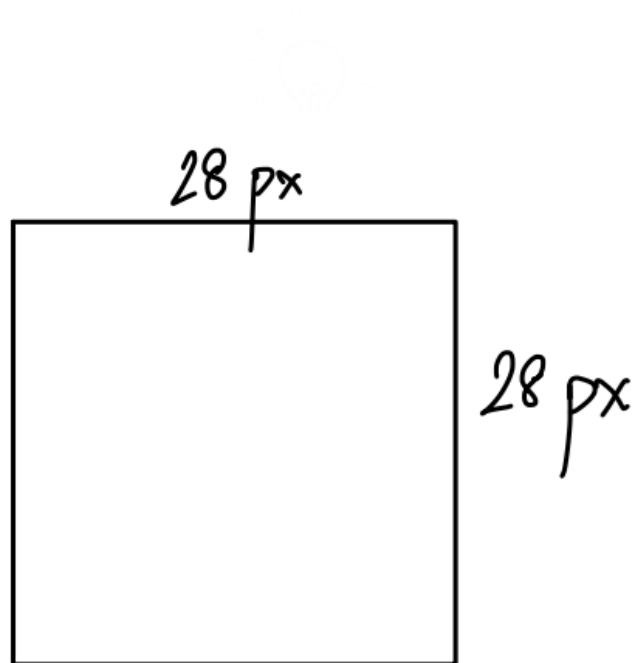
- **Learning** : Acquire data of create rules : Algorithms
- **Reasonings** : Choosing the right algorithm
- **Self Correction**: Five time algorithms to ensure most accurate result

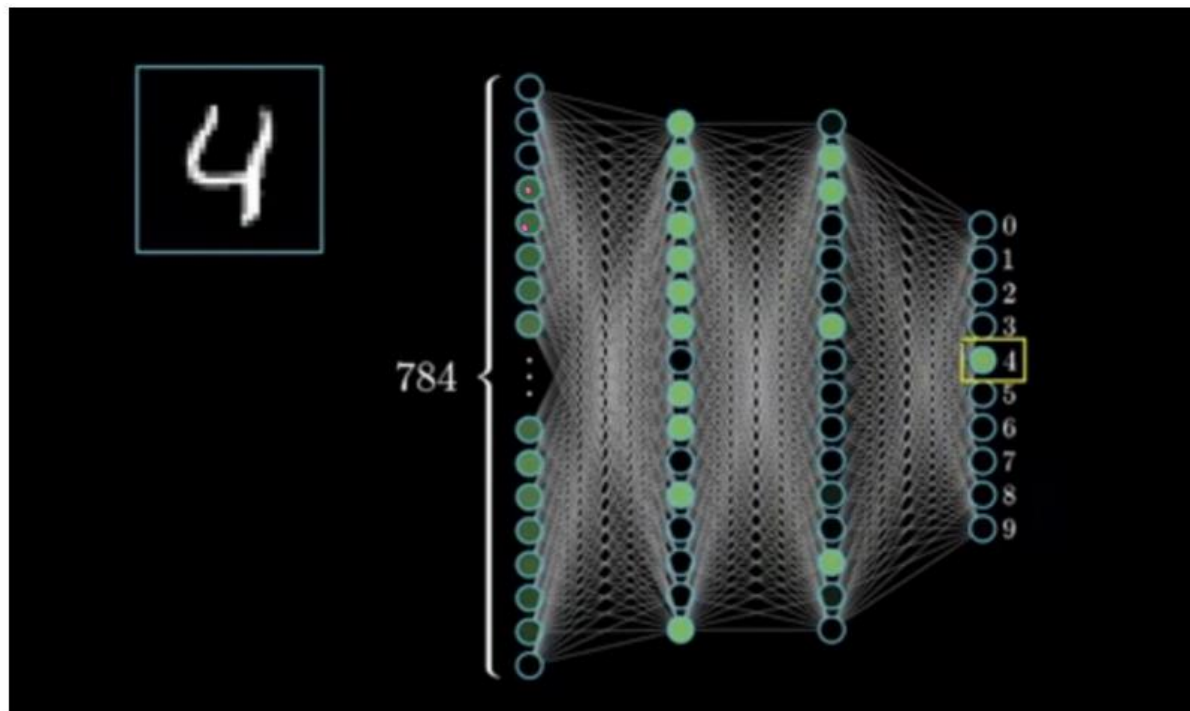
Artificial Intelligence

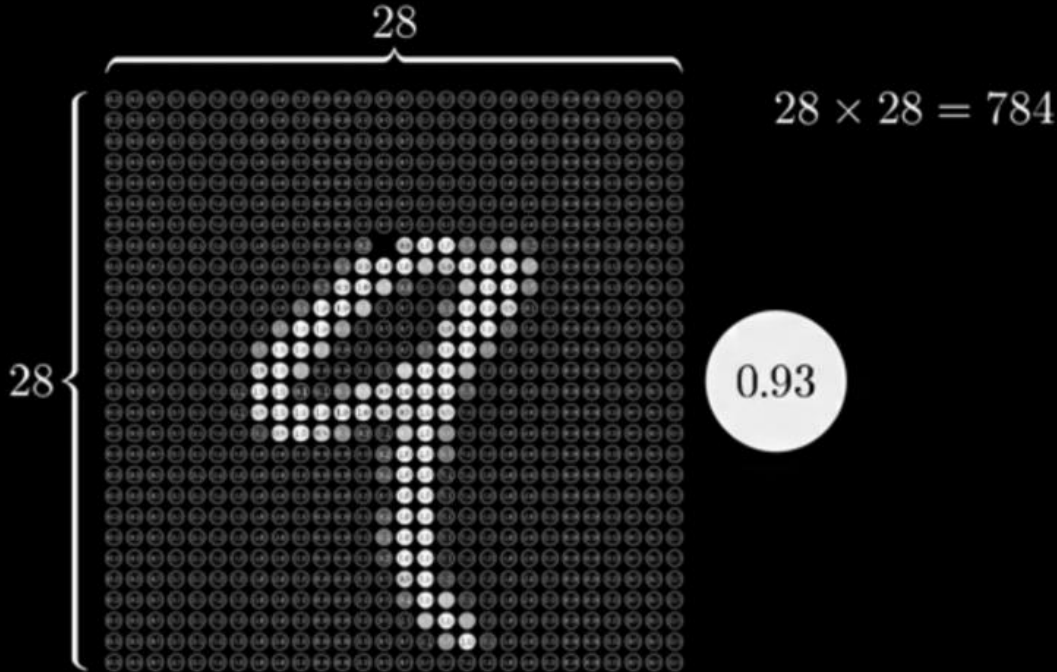


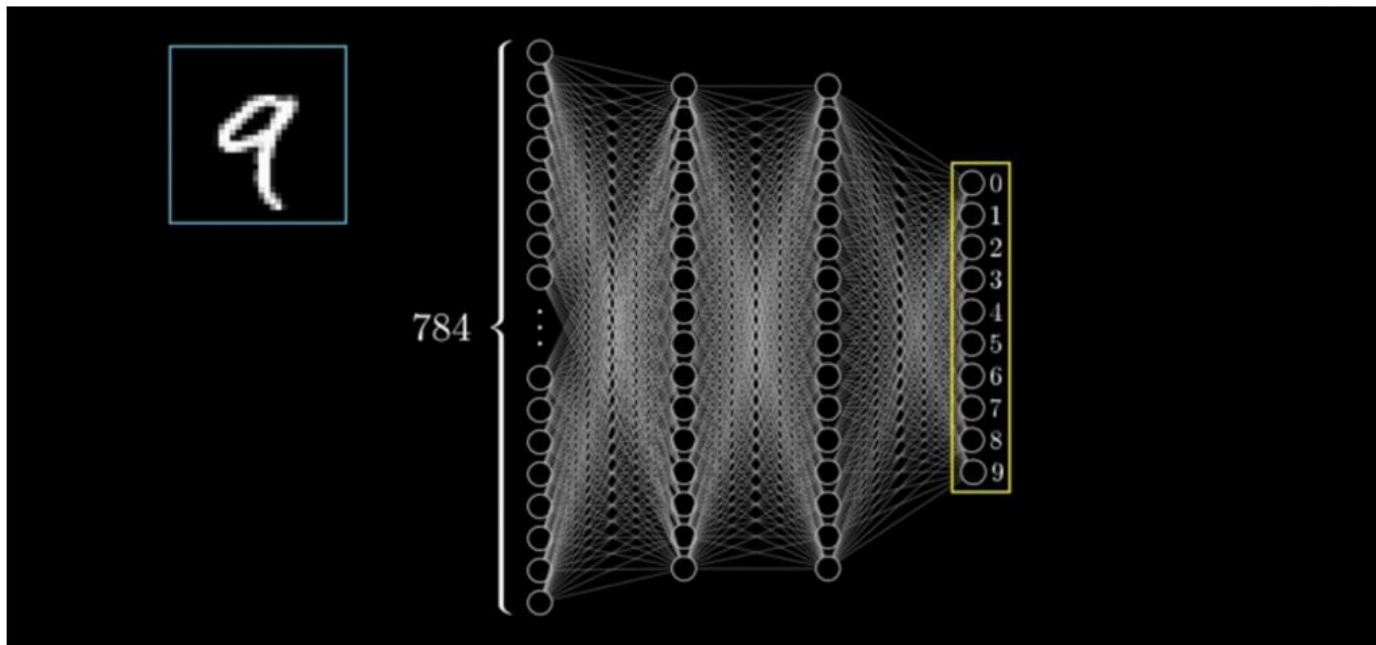
Artificial Intelligence

- **Understanding Intelligence**
 - **Neural Network**
 - **Inspired by the brains.**
 - **Computer learns the way we do.**

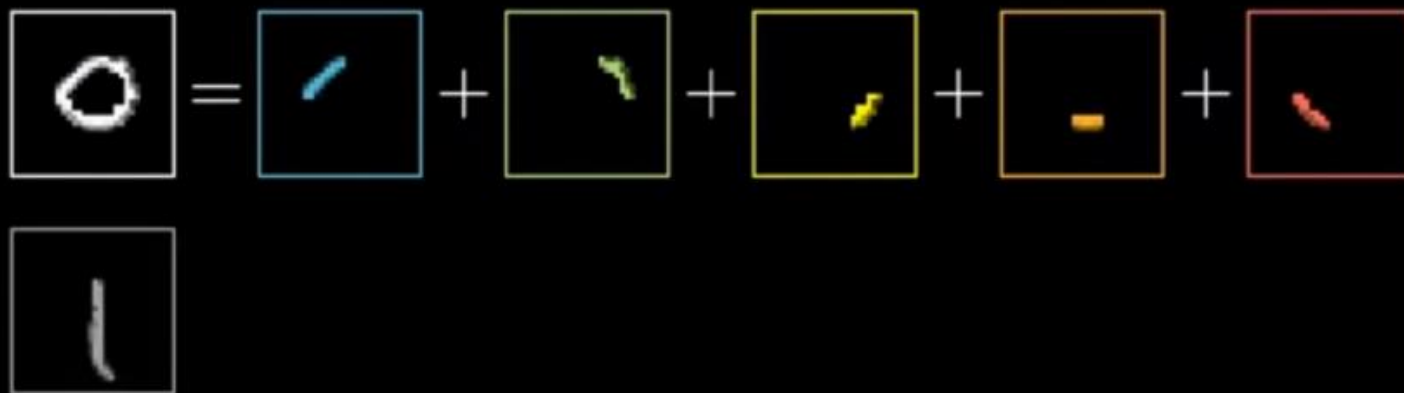


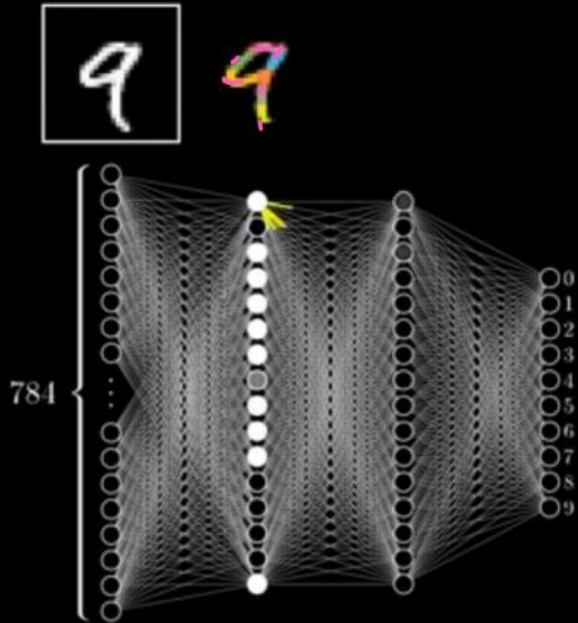


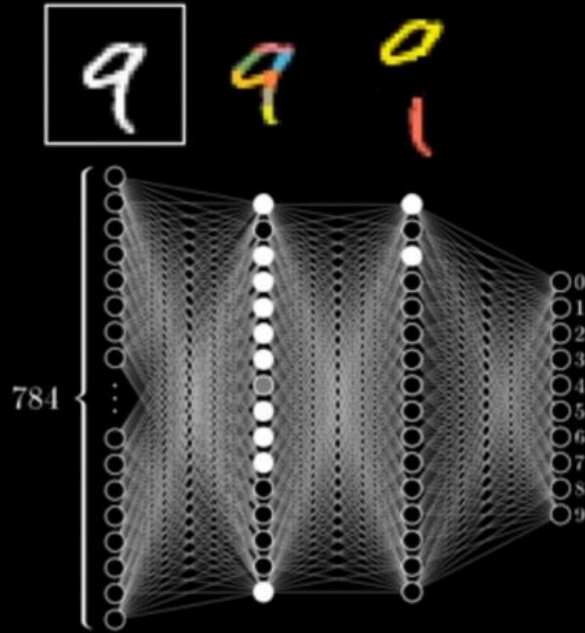




$$\begin{array}{l}
 9 = 0 + 1 \\
 8 = 0 + 0 \\
 4 = 1 + 1 + -
 \end{array}$$







Artificial Intelligence

AI TECHNOLOGOES

ILLUSTRATIVE SOLUTIONS



Sense

Computer Vision

Audio Processing



Comprehend

Natural Language Processing

Knowledge Representation



Act

Machine Learning

Expert System



Virtual Agents

Identity Analytics



Cognitive Robotics

Speech Analytics



Recommendation System

Data Visualization



Generative AI



- Type of artificial intelligence that focuses on creating new and original content that is indistinguishable from human-created content
- Uses algorithms and machine learning techniques to generate everything from images and music to text and video
- Trained on large datasets of existing content, such as images, text, music and videos

Type of Generative AI frameworks



- Generative Adversarial Networks (GANs): Uses two neural networks generator and a discriminator. The generator produces new data based on a given set of data, while the discriminator evaluates the data, determining if it is real or generated.
- VAEs: Uses an encoder-decoder architecture. The encoder encodes input data into a latent space, and the decoder decodes it back into the original data.
- Transformer Based Models: Read words and sentences in books, or articles. They learn from things they read and use that knowledge to create new things, like stories or poems.

Transformer Based Models (TBM)



- Uses a deep neural network.
- Trained on large dataset, allowing it to generate human like text or videos
- The transformer model consists of an encoder, which reads the input sequence of words and produces a representation of it, and a decoder, which produces the output sequence of words.
- Used for a wide range of tasks, question answering, text summarization, writing poems and stories.
- Programs based on TBM: OpenAI ChatGPT, Google BARD AI, Dall-E
- **Uses natural language processing (NLP) to understand and generate text data**

Transformer Based Models (TBM)



- Many words are synonyms of each other.
- **To understand words that are similar, to remove words that don't carry a substantive meaning and to simplify sentences this model uses NLP.**
- **So NLP helps the model be trained better.**

Natural Language Processing (NLP)



- Used to analyze and understand natural language text data
- Used by OpenAI ChatGPT, Google Bard AI to understand the data
- Steps involved:
 - Segmentation/ Tokenization: Breaking up a sentence or text into individual units, such as words, numbers. Each unit is referred to as a token.

0 1 2 3 4 5 6
I am learning about ChatGPT right now

Natural Language Processing (NLP)



- Removing Stop words: Removing words that don't contribute to the meaning of the sentence
This way it understands that 2 sentences have the same meaning for better pattern analysis.

0	1	2	3	4	5	6
i	am	learning	about	chatgpt	right	now

Natural Language Processing (NLP) - Cont'd



- Stemming: Removing the suffix or prefix of a word to obtain its root form.

0	1	2	3	4
i	am	learning	chatgpt	now

Natural Language Processing (NLP) - Cont'd



- Lemmatization: Normalizing a word to its base form, such as mapping a verb to its infinitive form.

0	1	2	3	4
i	be	learn	chatgpt	now

Natural Language Processing (NLP) - Cont'd



- Part of Speech Tagging: Assigning a part of speech to each token, such as noun, verb, adverb, etc.

0	1	2	3	4
i	be	learn	chatgpt	now
<i>proun</i>	<i>auxillary verb</i>	<i>verb</i>	<i>noun</i>	<i>adverb</i>

Natural Language Processing (NLP) - Cont'd



- Named Entity Recognition: Identifying and labeling entities, such as people, organizations, locations, dates, etc.

Ousted **WeWork** founder **Adam Neumann** lists his **Manhattan** penthouse for **\$37.5 million**

[organization] [person] [location] [monetary value]

CHAT GPT

- **Chat Generative Pre-trained Transformer** is an artificial intelligence chatbot developed by OpenAI and launched in November 2022.
- OpenAI has created ChatGPT, a start-up focused on artificial intelligence and its potential use cases.
- According to OpenAI's description, ChatGPT can answer “follow-up questions”, and can also “admit its mistakes, challenge incorrect premises, and reject inappropriate requests.”

CHAT GPT

- It is based on the company's GPT 3.5 series of **Language Learning Models (LLM)**.
- GPT stands for **Generative Pre-trained Transformer 3** and this is a kind of *computer language model* that relies on **deep learning techniques** to produce human-like text based on inputs.
- The model is trained to predict what will come next, and that's why one can technically have a 'conversation' with ChatGPT.

CHAT GPT

- chatbot was also trained using “**Reinforcement Learning from Human Feedback (RLHF)**.”
- Interestingly, ChatGPT has been trained to decline ‘inappropriate’ requests, presumably those which are ‘illegal’ in nature. However, it should be noted that ChatGPT has limitations, as it may generate incorrect information, and create “biased content.” More importantly, the chatbot’s knowledge of the world and events after 2021 is limited.
- Other products of Open AI
 - **Dall-E** — the AI-based text-to-image generator

CHAT GPT

- Google had recently showcased how its **LaMDA chatbot** is being used to help with fiction writing
 - LaMDA is a machine learning language model created by Google as a chatbot that is supposed to mimic humans in conversations
 - LaMBDA is built on neural network architecture

Human-AI Collaboration

Supportive Mode	Reimagining the Business	Ai Replaces the Humans
AI performs alongside humans by facilitating human judgments by providing resources such as predictive outputs.	AI performance activities that go beyond the cognitive abilities of humans. For e.g., large-scale genome study in bioinformatics.	Useful in potentially harmful situations for humans such as environments and rapid system response in nuclear reactors.

Benefits and applications of AI

Healthcare

- Address issues particularly in rural areas that suffer from poor connectivity and limited supply of healthcare professional .
- This can be achieved through AI driven diagnostics, early identification of potential pandemics, and imaging diagnostics.

Education and skilling

- Solve problem of quality and access issues observed in the Indian education sector.
- Augmenting and enhancing the learning experience through personalised learning, automating and expediting administrative tasks.

Benefits and applications of AI

Agriculture

- Driving a food revolution to meet the increased demand for food by improvement in crop yield.
- Address challenges in irrigation, misuse of pesticides and fertilisers through real time advisory.

Manufacturing

- Manufacturing industry is expected to be one of the biggest beneficiaries of AI based solutions in field of engineering, supply chain management, production and quality assurance etc.

Benefits and applications of AI

Energy

- AI can enable storage of energy through intelligent grids enable by smart meters, and also improve the reliability and affordability of photovoltaic energy.
- AI may also be deployed for predictive maintenance of grid infrastructure.

Financial Services

- AI can ensure early detection of financial risk and systemic failures, and automation of reduce malicious intent in financial system, such as market manipulation, anomalous trading etc.

Benefits and applications of AI

Defence and Security

- AI can be used for intelligence gathering, cyber defence, risk terrain analysis, detecting anomalous behaviour in human etc.

Law Enforcement

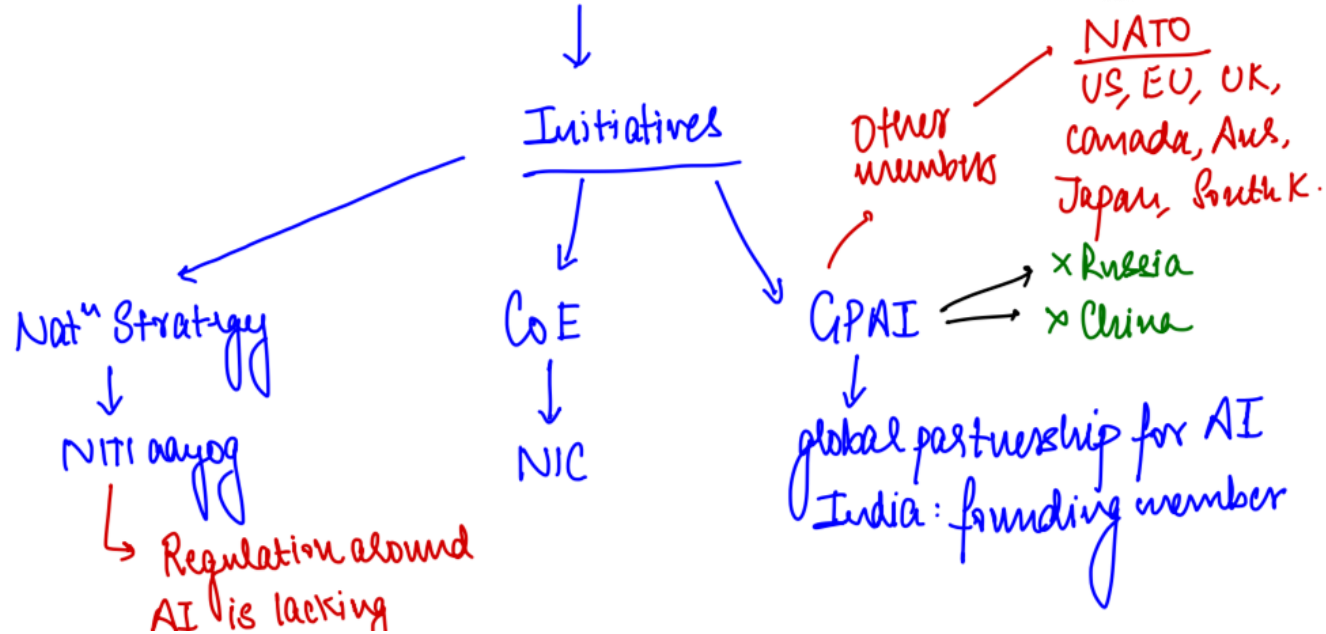
- AI technologies can be used by law enforcement which include facial recognition, speech recognition, predictive analytics etc.

Disaster Management

- AI has potential in providing remedial measures and control in aftermath of man-made recognition can be used in infrastructure damage assessments.

Issue	
General Context	● Indian Context ● Awareness regulation ● Funding ● HR ● Infra.

Indian Scenario



Introducing Bard, an experimental conversational AI service powered by LaMDA

You can use Bard to —

Plan a friend's baby shower

Compare two Oscar nominated movies

Get lunch ideas based on what's in your fridge

What's on your mind?



Bard may give inaccurate or inappropriate information. Your feedback makes Bard more helpful and safe.

E-SIM

- an eSIM is an embedded SIM
 - It has same hardware of a regular SIM card chip
 - It is a permanently embedded part of the motherboard of a device.
- e-SIM acting as a unique identifier for telecom operators to reach the exact smartphone when they make a call or send a text.
- eSIMs were first established a decade ago in 2012.

E-SIM

Benefits

- users switch operators without having to replace any physical SIM cards.
- **Convenience:** The ability to store multiple SIM profiles in your eSIM also means you can switch between profiles easily, without the need of activating a SIM repeatedly or physically switching cards repeatedly.

E-SIM

Benefits

- **Security:** In case of theft of phone, SIM card can not be misused as it is embedded part of smart phone.
- **One less opening on your phone:** Having an eSIM also means there is one less opening on the frame of your phone, in theory, which should reduce the likelihood of elements like dust and water entering the phone from yet another slot.

E-SIM

Disadvantages

- **Emergencies:** If your phone stops working, runs out of battery or simply, falls and gets a cracked screen, your communication is brought to a complete standstill with eSIMs.
- **Unusable in countries with no eSIM support:** You cannot use an eSIM phone in a country where the telecom operators simply don't support the technology yet.
- **Support only available in premium phones:** In India, eSIM support is currently available on more expensive devices.

D2M Technology

- Direct To Mobile Tech—>broadcast video and other forms of multimedia content directly to mobile phones
- It does not require an active internet connection

D2M Technology

Concept

- D2M is based on the convergence of broadband and broadcast, using which mobile phones can receive terrestrial digital TV.
- It would be similar to how people listen to FM radio on their phones, where a receiver within the phone can tap into radio frequencies.
- Using D2M, multimedia content can also be beamed to phones directly.

D2M Technology

Benefits

- It can be used to directly broadcast content related to citizen-centric information
- It can be used to counter fake news, issue emergency alerts and offer assistance in disaster management
- It can be used to broadcast live news, sports etc. on mobile phones.

D2M Technology

Impact

- Consumers
 - Can access multimedia content from Video on Demand (VoD) or Over The Top (OTT) content platforms without having to exhaust their mobile data at a nominal rate.
 - The technology will also allow people from rural areas, with limited or no internet access, to watch video content.

D2M Technology

Businesses

- It can enable telecom service providers to offload video traffic from their mobile network onto the broadcast network, thus helping them to **decongest valuable mobile spectrum**.
- This will also **improve usage of mobile spectrum** and free up bandwidth which will help reduce call drops, increase data speeds etc.

D2M Technology

Government Measures

- The Department of Telecommunications (DoT) has set up a committee to study the feasibility of a spectrum band for offering broadcast services directly to users' smartphones

D2M Technology

Challenges

- bringing key stakeholders like mobile operators onboard will be the “biggest challenge” in launching D2M technology on a wide scale.
- mass roll out of the technology will entail changes in infrastructure and some regulatory changes.

CrysXPP

- Researchers from **IIT Kharagpur** in collaboration with the **Indo-Korea Science and Technology Center (IKST)** have developed a method called **CrysXPP**.
- It is a **machine learning (ML) system** that enables rapid prediction of various **material properties** with high precision.

CrysXPP

- ML is a type of artificial intelligence (AI) that allows software applications to become more accurate at predicting outcomes without being explicitly programmed to do so
- Machine learning algorithms use historical data as input to predict new output values
- Applications of ML: Recommendation engines, fraud detection, spam filtering, malware threat detection, business process automation (BPA) and Predictive maintenance.

CrysXPP

Applications

- **CrysXPP made large-scale testing easier**, because until now, it has been difficult to test crystalline materials on a large scale.
- It also helps determine the electronic, **magnetic and elastic properties of crystals**, which are often time- consuming, expensive and computation-intensive.

Web 5.0

- Former Twitter CEO recently announced his vision for a new decentralized web platform that is being called Web 5.0.
- Objective—> “ownership of data and identity to individuals”.
- It is a **Decentralized Web Platform** that enables developers to leverage Decentralized Identifiers, Verifiable Credentials, and Decentralized Web Nodes to write Decentralized Web Apps.
- Web 5.0 is Web 2.0 plus Web 3.0 that will allow users to own their identity on the Internet and control their data.
- It aimed at building an extra decentralized web that puts one in control of your data and identity.

Web 5.0

Web 1.0 : Read only web

Web 2.0 : Social Web

Web 3.0 : Semantic Web

Web 5.0

Web 4.0 (Mobile Web)

- It is already a work in progress, with no clear understanding of what it would entail.
- It is often referred to as the **symbiotic web**.
 - It will make contact between humans and computers in symbiosis.

Geospatial Data

- **The Ministry of Science and Technology** has notified a citizen-centric **National Geospatial Policy (NGP) 2022**.
- **Geospatial Data Promotion and Development Committee (GDPDC)**: At national level, an apex body for formulating and implementing appropriate guidelines, strategies, and programs.
 - **GDPDC would replace the National** Spatial Data Committee (NSDC) and Geospatial Data Promotion and Development Committee.

Geospatial Data

Geospatial Data Infrastructure

- GDPDC will **develop 14 global geospatial data themes** recognized by the United Nations Statistics Division.
- **National Geospatial Data Registry (NGDR):** Accessible by all stakeholders.
- **Unified Geospatial Interface (UGI):** To provide consumer-oriented products and solutions using Geospatial data.
- **Survey of India (SoI):** Will help in developing NGDR and UGI in collaboration with Bhask`atics (BISAG-N)-under MeitY, private sector etc.

Geospatial Data

- **Mapping infrastructure:** Policy shall replace the National Map Policy, 2005.
- **Geospatial Knowledge Infrastructure (GKI):** Will be enabled by the integration of Geospatial data with Fourth Industrial Revolution technologies and the growing digital infrastructure.
- **Geospatial Education and Skill Development:** National Institute for Geo-informatics Science and Technology (NIGST), Indian Institute of Remote Sensing (IIRS), and/or any suitable institute(s), public or private, will be developed into Centre(s) of Excellence.
- **Geospatial Industrial Development Board (GIDB):** Advisory body under GDPDC.

National Data Governance Policy

Aim—>

- To **enhance access, quality, and use of data**
- It will **provide datasets access to Indian researchers and startups.**
- **NDGFP does not apply to the private players**, but they can voluntarily contribute their datasets.
- No provision for monetization of such data.

National Data Governance Policy

Components

- **Indian Datasets Platform:** Consisting of anonymized non-personal datasets from Government entities collected from Indian citizens or those in India
- **Indian Data Management Office (IDMO)**
 - It will be set up under Digital India Corporation (DIC), under MeitY
- **Data Management Units (DMUs):** In every Ministry/Department to work closely with IDMO for ensuring implementation of the Policy

National Data Governance Policy

Objectives of the NDGFP are:

- To accelerate Digital Governance.
- To have standardized data management and security standards across whole of Government;
- To accelerate creation of common standard based public digital platforms while ensuring privacy, safety, and trust; (d) To have standard APIs and other tech standards for Whole of Government Data management and access.

National Data Governance Policy

Objectives of the NDGFP are:

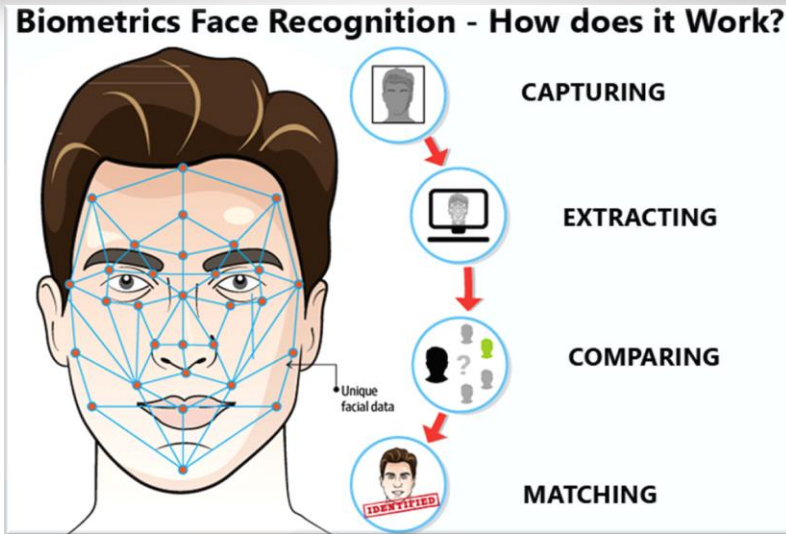
- To promote transparency, accountability, and ownership in Non-personal data and Datasets access. For purposes of safety and trust, any non-personal data sharing by any entity can be only via platforms designated and authorised by IDMO. (1) To build a platform that will allow Dataset requests to be received and processed.

National Data Governance Policy

Objectives of the NDGFP are:

- To build Digital Government goals and capacity, knowledge and competency in Government departments and entities.
- To set quality standards and promote expansion of India Datasets program and overall non-personal Datasets Ecosystem.
- To ensure greater citizen awareness, participation, and engagement.

Facial Recognition System



Facial Recognition System

- FRS is a way of identifying or confirming an individual's identity using their face.
- It can be used to identify people in photos, videos, or in real-time.
- Computer algorithms map unique **facial-landmarks** such as shape of cheekbones, contours of lips etc. and convert these into a numerical code— termed a **faceprint**.

Facial Recognition System

Applications

- FRS is part of the **Digi Yatra initiative** to promote seamless and hassle-free experience at airports and simultaneously improving the security.
- Government is looking to make ticket booking, airport entry and boarding pass security check-in digital.

DIFFERENT TYPES OF BIOMETRICS:



Typing Style



Navigation Style

Behavioral
Biometric Identifiers:



Interaction Style



Physical Style



Signature



Face Recognition

Physical
Biometric Identifiers:



Eye Scanners



Fingerprints



Voice Recognition



DNA

Chemical & Vein
Biometric Identifiers:



Vein Recognition

India Stack

- India stack is a collection of open APIs (**Application Programming Interface**) and digital public goods
- IndiaStack includes **APIs of Aadhaar, Unified Payment Interface (UPI), Co-Win, DigiLocker, Aarogya Setu, eSanjeevani, UMANG, DIKSHA**, etc
- Its objective is to unlock the economic opportunities of **identity, data, and payments** at population scale.

Remaining Terms

- **Digital Signature**
- **Software as a service**

Test of 20 Questions!

Cut off : 15/20



Q.1 (2016)

Q. With reference to 'LiFi', recently in the news, which of the following statements is/are correct?

1. It uses light as the medium for high-speed data transmission.
2. It is a wireless technology and is many times faster than 'WiFi'.

Select the correct answer using the code given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.2 (2016)

Q. 'Project Loon', sometimes seen in the news, is related to

- a) Waste management technology
- b) Wireless communication technology
- c) Solar power production technology
- d) Water conservation technology

Q.3 (2018)

Q. Consider the following pairs :

Terms sometimes seen in news

Context /Topic

1. Belle II experiment -

Artificial Intelligence

2. Blockchain technology -

Digital/Cryptocurrency

3. CRISPR — Cas9 -

Particle Physics

Which of the pairs given above is/are correctly matched ?

a) 1 and 3 only

b) 2 only

c) 2 and 3 only

d) 1, 2 and 3

Q.4 (2018)

Q. When the alarm of your smartphone rings in the morning, you wake up and tap it to stop the alarm which causes your geyser to be switched on automatically. The smart mirror in your bathroom shows the day's weather and also indicates the level of water in your overhead tank. After you take some groceries from your refrigerator for making breakfast, it recognises the shortage of stock in it and places an order for the supply of fresh grocery items. When you step out of your house and lock the door, all lights, fans, geysers and AC machines get switched off automatically. On your way to the office, your car warns you about traffic congestion ahead and suggests an alternative route, and if you are late for a meeting, it sends a- message to your office accordingly.

In the context of emerging communication technologies, which one of the following term" best applies to the above scenario?

- a) Border Gateway Protocol
- b) Internet of Things
- c) Internet Protocol
- d) Virtual Private Network

Q.5 (2018)

Q. "3D printing" has applications in which of the following?

1. Preparation of confectionery items
2. Manufacture of bionic ears
3. Automotive industry
4. Reconstructive surgeries
5. Data processing technologies

Select the correct answer using the code given below :

- | | |
|--------------------|---------------------|
| a) 1, 3 and 4 only | b) 2, 3 and 5 only |
| c) 1 and 4 only | d) 1, 2, 3, 4 and 5 |

Q.6 (2019)

Q. With reference to communication technologies, what is/are the difference/differences between LTE (Long-Term Evolution) and VoLTE (Voice over Long-Term Evolution)?

1. LTE is commonly marketed as 3G and VoLTE is commonly marketed as advanced 3G.
2. LTE is data-only technology and VoLTE is voice-only technology.

Select the correct answer using the code given below.

- | | |
|-----------------|--------------------|
| a) 1 only | b) 2 only |
| c) Both 1 and 2 | d) Neither 1 nor 2 |

Q.7 (2019)

Q. In the context of digital technologies for entertainment, consider the following statements:

1. In Augmented Reality (AR), a simulated environment is created and the physical world is completely shut out.
2. In Virtual Reality (VR), images generated from a computer are projected into real-life objects or surroundings.
3. AR allows individuals to be present in the world and improves the experience using the camera of smart-phone or PC.
4. VR closes the world, and transposes an individual, providing complete immersion experience.

Which of the statements given above is/are correct?

- | | |
|-----------------|------------|
| a) 1 and 2 only | b) 3 and 4 |
| c) 1, 2 and 3 | d) 4 only |

Q.8 (2019)

Q. Consider the following statements: A digital signature is

1. an electronic record that identifies the certifying authority issuing it
2. used to serve as a proof of identity of an individual to access information or server on Internet
3. an electronic method of signing an electronic document and ensuring that the original content is unchanged

Which of the statements given above is/are correct?

- | | |
|-----------|-----------------|
| a) 1 only | b) 2 and 3 only |
| c) 3 only | d) 1, 2 and 3 |

Q.9 (2019)

Q. In the context of wearable technology, which of the following tasks is/are accomplished by wearable devices?

1. Location identification of a person
2. Sleep monitoring of a person
3. Assisting the hearing impaired person

Select the correct answer using the code given below:

- a) 1 only
- b) 2 and 3 only
- c) 3 only
- d) 1, 2 and 3

Q.10 (2020)

Q. Consider the following activities:

1. Spraying pesticides on a crop field
 2. Inspecting the craters of active volcanoes
 3. Collecting breath samples from spouting whales for DNA analysis
- At the present level of technology, which of the above activities can be successfully carried out by using drones?
- a) 1 and 2 only
 - b) 2 and 3 only
 - c) 1 and 3 only
 - d) 1,2 and 3

Q.11 (2020)

Q. In India, the term “Public Key Infrastructure” is used in the context of

- a) Digital security infrastructure
- b) Food security infrastructure
- c) Health care and education infrastructure
- d) Telecommunication and transportation infrastructure

Q.12 (2020)

Q. With the present state of development, Artificial Intelligence can effectively do which of the following?

1. Bring down electricity consumption in industrial units.
2. Create meaningful short stories and songs.
3. Disease diagnosis.
4. Text-to-Speech Conversion.
5. Wireless transmission of electrical energy.

Select the correct answer using the code given below:

- | | |
|-----------------------|---------------------|
| a) 1, 2, 3 and 5 only | b) 1, 3 and 4 only |
| c) 2, 4 and 5 only | d) 1, 2, 3, 4 and 5 |

Q.13 (2020)

Q. With reference to Visible Light Communication (VLC) technology, which of the following statements are correct?

1. VLC uses electromagnetic spectrum wavelengths 375 to 780 nm.
2. VLC is known as long-range optical wireless communication.
3. VLC can transmit large amounts of data faster than Bluetooth
4. VLC has no electromagnetic interference.

Select the correct answer using the code given below:

- a) 1, 2 and 3 only
- b) 1, 2 and 4 only
- c) 1, 3 and 4 only
- d) 2, 3 and 4 onl

Q.14 (2020)

Q. With reference to “Blockchain Technology”, consider the following statements:

1. It is a public ledger that everyone can inspect, but which no single user controls.
2. The structure and design of blockchain is such that all the data in it are about cryptocurrency only.
3. Applications that depend on basic features of blockchain can be developed without anybody’s permission.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 only
- d) 1 and 3 only

Q.15 (2022)

Q. With reference to Web 3.0, consider the following statements:

1. Web 3.0 technology enables people to control their own data.
2. In Web 3.0 world, there can be blockchain-based social networks.
3. Web 3.0 is operated by users collectively rather than a corporation

Which of the following given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Q.16 (2022)

Q. With reference to “Software as a Service”, Consider the following statements.

1. SaaS buyers can customise the user interface and can change data fields.
2. SaaS users can access their data through their mobile devices.
3. Outlook, Hotmail and Yahoo! Mail are forms of SaaS.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Q.17 (2022)

Q. Which one of the following is the context in which the term “qubit” is mentioned?

- (a) Cloud Services
- (b) Quantum Computing
- (c) Visible Light Communication Technologies
- (d) Wireless Communication Technologies

Q.18 (2022)

Q. Consider the following communication technologies:

1. Closed-circuit Television
2. Radio Frequency Identification
3. Wireless Local Area Network

Which of the above are considered of Short-Range devices/technologies?

- (a) 1 and 2 only.
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Q.19 (2014)

Q. In addition to fingerprint scanning, which of the following can be used in the biometric identification of a person?

1. Iris scanning
2. Retinal scanning
3. Voice recognition

Select the correct answer using the code given below.

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.20 (2015)

Q. With reference to 'Near Field Communication (NFC) Technology', which of the following statements is/are correct?

1. It is a contactless communication technology that uses electromagnetic radio fields
2. NFC is designed for use by devices which can be at a distance of even a metre from each other
3. NFC can use encryption when sending sensitive information.

Select the correct answer using the code given below.

- | | |
|-----------------|--------------|
| a) 1 and 2 only | b) 3 only |
| c) 1 and 3 only | d) 1,2 and 3 |

Congratulations on giving the test!

Let's Discuss The Answers.



Q.1 (2016)

Q. With reference to 'LiFi', recently in the news, which of the following statements is/are correct?

1. It uses light as the medium for high-speed data transmission.
2. It is a wireless technology and is many times faster than 'WiFi'.

Select the correct answer using the code given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2**
- d) Neither 1 nor 2

Answer: C

- LiFi is a wireless optical networking technology that uses light-emitting diodes (LEDs) for data transmission.
- LiFi is designed to use LED light bulbs similar to those currently in use in many energy-conscious homes and offices. However, LiFi bulbs are outfitted with a chip that modulates the light imperceptibly for optical data transmission.
- LiFi data is transmitted by the LED bulbs and received by photoreceptors.
- In terms of its end use the technology is similar to Wi-Fi.
- The key technical difference is that Wi-Fi uses radio frequency to transmit data.
- Using light to transmit data allows Li-Fi to offer several advantages like working across higher bandwidth, working in areas susceptible to electromagnetic interference (e.g. aircra

Q.2 (2016)

Q. 'Project Loon', sometimes seen in the news, is related to

- a) Waste management technology
- b) Wireless communication technology**
- c) Solar power production technology
- d) Water conservation technology

Answer: B

- Loon LLC is an Alphabet Inc.(Parent company of Google) subsidiary working on providing Internet access to rural and remote areas.
- The company uses high-altitude balloons placed in the stratosphere at an altitude of about 18 km (11 mi) to create an aerial wireless network with up to 4G-LTE speeds.
- It was named Project Loon since even Google itself found the idea of providing Internet access to the remaining 5 billion population unprecedented and "loony".
- It may also be a reference to the balloons used.

Q.3 (2018)

Q. Consider the following pairs :

Terms sometimes seen in news

Context /Topic

1. Belle II experiment -

Artificial Intelligence

2. Blockchain technology -

Digital/Cryptocurrency

3. CRISPR — Cas9 -

Particle Physics

Which of the pairs given above is/are correctly matched ?

a) 1 and 3 only

b) 2 only

c) 2 and 3 only

d) 1, 2 and 3

Answer: B

- The blockchain is an undeniably ingenious invention which created the backbone of a new type of internet originally devised for the digital currency, Bitcoin.
- Genome editing (also called gene editing) is a group of technologies that give scientists the ability to change an organism's DNA. These technologies allow genetic material to be added, removed, or altered at particular locations in the genome.
- Several approaches to genome editing have been developed.
- A recent one is known as CRISPR-Cas9, which is short for clustered regularly interspaced short palindromic repeats and CRISPR-associated protein 9.
- The CRISPR-Cas9 system has generated a lot of excitement in the scientific community because it is faster, cheaper, more accurate, and more efficient than other existing genome editing methods.

Q.4 (2018)

Q. When the alarm of your smartphone rings in the morning, you wake up and tap it to stop the alarm which causes your geyser to be switched on automatically. The smart mirror in your bathroom shows the day's weather and also indicates the level of water in your overhead tank. After you take some groceries from your refrigerator for making breakfast, it recognises the shortage of stock in it and places an order for the supply of fresh grocery items. When you step out of your house and lock the door, all lights, fans, geysers and AC machines get switched off automatically. On your way to the office, your car warns you about traffic congestion ahead and suggests an alternative route, and if you are late for a meeting, it sends a- message to your office accordingly.

In the context of emerging communication technologies, which one of the following term" best applies to the above scenario?

- a) Border Gateway Protocol
- c) Internet Protocol

b) Internet of Things

- d) Virtual Private Network

Answer: B

- Internet of things is a seamless connected network system of embedded objects/ devices, with identifiers, in which communication without any human intervention is possible using standard and interoperable communication protocols. With this technology, each and every device that we use in daily life (including refrigerators, cars and even toothbrush) can be made intelligent and smart by using device-specific operating systems and connectivity to the Internet

Q.5 (2018)

Q. "3D printing" has applications in which of the following?

1. Preparation of confectionery items
2. Manufacture of bionic ears
3. Automotive industry
4. Reconstructive surgeries
5. Data processing technologies

Select the correct answer using the code given below :

a) 1, 3 and 4 only

b) 2, 3 and 5 only

c) 1 and 4 only

d) 1, 2, 3, 4 and 5

Answer: D

- There are endless possibilities for the utilization of 3d Printing. We can do many things with 3d printing machines. All the above things can be done with the help of 3D Printing

Q.6 (2019)

Q. With reference to communication technologies, what is/are the difference/differences between LTE (Long-Term Evolution) and VoLTE (Voice over Long-Term Evolution)?

1. LTE is commonly marketed as 3G and VoLTE is commonly marketed as advanced 3G.
2. LTE is data-only technology and VoLTE is voice-only technology.

Select the correct answer using the code given below.

- | | |
|-----------------|---------------------------|
| a) 1 only | b) 2 only |
| c) Both 1 and 2 | d) Neither 1 nor 2 |

Answer: D

LTE and VoLTE

- VoLTE is an HD voice calling Service over 4G LTE rather than 2G/3G network present we are using.
- VoLTE is a technology wherein you can simultaneously send voice and data over the network without diminishing quality of voice. In the case of LTE, if you make a voice call and you also keep your data connection on, the quality of voice will reduce.
- So, in order to make a good quality voice call, you will have to switch off data. You may have noticed that in case of 2G and 3G, while you are making a phone call, some phones will automatically stop data transfer so as to preserve the voice quality.
- But in case of VoLTE, the voice quality will not reduce even if your data connection is on. With VoLTE it is very easy to transmit telephone conversation over the data network.

Q.7 (2019)

Q. In the context of digital technologies for entertainment, consider the following statements:

1. In Augmented Reality (AR), a simulated environment is created and the physical world is completely shut out.
2. In Virtual Reality (VR), images generated from a computer are projected into real-life objects or surroundings.
3. AR allows individuals to be present in the world and improves the experience using the camera of smart-phone or PC.
4. VR closes the world, and transposes an individual, providing complete immersion experience.

Which of the statements given above is/are correct?

- a) 1 and 2 only **b) 3 and 4**
c) 1, 2 and 3 d) 4 only

Answer: B

- Augmented reality (AR) is an interactive experience of a real-world environment where the objects that reside in the real-world are enhanced by computer-generated perceptual information, sometimes across multiple sensory modalities, including visual, auditory, haptic, somatosensory and olfactory
- The origin of the word augmented is augment, which means to add or enhance something. In the case of Augmented Reality (also called AR), graphics, sounds, and touch feedback are added into our natural world to create an enhanced user experience.

- Cell phones apps and business applications by companies using augmented reality are a few of the many applications driving augmented reality application development.
- The key point is that the information provided is highly topical and relevant to what you want you are doing.
- Virtual reality (VR) is an experience taking place within simulated and immersive environments that can be similar to or completely different from the real world.
- Applications of virtual reality can include entertainment (i.e. gaming) and educational purposes (i.e. medical or military training). Other, distinct types of VR style technology include augmented reality and mixed reality.
- 1st and 2nd are wrong statement while, 3rd and 4th are the correct.

Q.8 (2019)

Q. Consider the following statements: A digital signature is

1. an electronic record that identifies the certifying authority issuing it
2. used to serve as a proof of identity of an individual to access information or server on Internet
3. an electronic method of signing an electronic document and ensuring that the original content is unchanged

Which of the statements given above is/are correct?

- | | |
|------------------|-----------------|
| a) 1 only | b) 2 and 3 only |
| c) 3 only | d) 1, 2 and 3 |

Answer: C

- A digital signature is basically a way to ensure that an electronic document (e-mail, spreadsheet, text file, etc.) is authentic. Authentic means that you know who created the document and you know that it has not been altered in any way since that person created it.
- Digital signatures rely on certain types of encryption to ensure authentication.
- Encryption is the process of taking all the data that one computer is sending to another and encoding it into a form that only the other computer will be able to decode.
- Authentication is the process of verifying that information is coming from a trusted source.
- These two processes work hand in hand for digital signatures.

Q.9 (2019)

Q. In the context of wearable technology, which of the following tasks is/are accomplished by wearable devices?

1. Location identification of a person
2. Sleep monitoring of a person
3. Assisting the hearing impaired person

Select the correct answer using the code given below:

- a) 1 only
- b) 2 and 3 only
- c) 3 only

d) 1, 2 and 3

Answer: D

- The terms “wearable technology”, “wearable devices”, and “wearables” all refer to electronic technologies or computers that are incorporated into items of clothing and accessories which can comfortably be worn on the body
- Examples of wearable devices include watches, glasses, contact lenses, e-textiles and smart fabrics, headbands, beanies and caps, jewelry such as rings, bracelets, and hearing aid-like devices that are designed to look like earrings.
- The implications and uses of wearable technology are far reaching and can influence the fields of health and medicine, fitness, aging, disabilities, education, transportation, enterprise, finance, gaming and music.

Q.10 (2020)

Q. Consider the following activities:

1. Spraying pesticides on a crop field
2. Inspecting the craters of active volcanoes
3. Collecting breath samples from spouting whales for DNA analysis

At the present level of technology, which of the above activities can be successfully carried out by using drones?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1,2 and 3**

Answer: D

- Without crop protection, including pesticides, more than half of the world's crops would be lost to insects, diseases and weeds. Pesticides are important.
- A crater is a bowl-shaped depression, or hollowed-out area, produced by the impact of a meteorite, volcanic activity, or an explosion. Craters produced by the collision of a meteorite with the Earth (or another planet or moon) are called impact craters.

Q.11 (2020)

Q. In India, the term “Public Key Infrastructure” is used in the context of

- a) Digital security infrastructure**
- b) Food security infrastructure
- c) Health care and education infrastructure
- d) Telecommunication and transportation infrastructure

Answer: A

- Public Key Infrastructure (PKI) is a technology for authenticating users and devices in the digital world. Under this system, one or more trusted parties digitally sign documents certifying that a particular cryptographic key belongs to a particular user or device. The key can then be used as an identity for the user in digital networks.
- Therefore, option (a) is the correct answer

Q.12 (2020)

Q. With the present state of development, Artificial Intelligence can effectively do which of the following?

1. Bring down electricity consumption in industrial units.
2. Create meaningful short stories and songs.
3. Disease diagnosis.
4. Text-to-Speech Conversion.
5. Wireless transmission of electrical energy.

Select the correct answer using the code given below:

- | | |
|-----------------------|---------------------------|
| a) 1, 2, 3 and 5 only | b) 1, 3 and 4 only |
| c) 2, 4 and 5 only | d) 1, 2, 3, 4 and 5 |

Answer: B

- Statement 1 is correct→Bring down electricity consumption in industrial units by using efficient usage of electricity supply by use of data analytics
- Statement 2 is incorrect→Creating meaningful short stories and songs is not possible by AI because of 'Meaningful' keyword in stories and songs can not be made by AI
- Statement 3 is correct→Disease diagnosis by Machine Learning models
- Statement 4 is correct→Text-to-Speech Conversion is possible by help of Natural Language Processing in AI
- Statement 4 is incorrect→Wireless transmission of electrical energy is unrelated to AI

Q.13 (2020)

Q. With reference to Visible Light Communication (VLC) technology, which of the following statements are correct?

1. VLC uses electromagnetic spectrum wavelengths 375 to 780 nm.
2. VLC is known as long-range optical wireless communication.
3. VLC can transmit large amounts of data faster than Bluetooth
4. VLC has no electromagnetic interference.

Select the correct answer using the code given below:

- a) 1, 2 and 3 only
- b) 1, 2 and 4 only
- c) 1, 3 and 4 only**
- d) 2, 3 and 4 onl

Answer: C

- Visible light communication (VLC) is a data communications variant which uses visible light between 400 and 800 THz (780–375 nm). VLC is a subset of optical wireless communications technologies. 1 is correct.
- The technology uses fluorescent lamps (ordinary lamps, not special communications devices) to transmit signals at 10 kbit/s, or LEDs for up to 500 Mbit/s over short distances.
- Systems such as RONSJA can transmit at full Ethernet speed (10 Mbit/s) over distances of 1–2 kilometres (0.6–1.2 mi). It is not long range. 2 is incorrect.
- VLC has the characteristic, to resist electromagnetic interference. 4 is correct.
- It carries more data than WiFi and bluetooth. 3 is correct.

Q.14 (2020)

Q. With reference to “Blockchain Technology”, consider the following statements:

1. It is a public ledger that everyone can inspect, but which no single user controls.
2. The structure and design of blockchain is such that all the data in it are about cryptocurrency only.
3. Applications that depend on basic features of blockchain can be developed without anybody’s permission.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 only

d) 1 and 3 only

Answer: D

- Blockchain technology is a structure that stores transactional records, also known as the block, of the public in several databases, known as the “chain,” in a network connected through peer-to-peer nodes. Typically, this storage is referred to as a ‘digital ledger.’
- In simpler words, the digital ledger is like a Google spreadsheet shared among numerous computers in a network, in which, the transactional records are stored based on actual purchases.
- The fascinating angle is that anybody can see the data, but they can’t corrupt it. 1 is correct.
- Blockchain can be used for many purposes like Secure sharing of medical data, Music royalties tracking. 2 is incorrect. It is an extreme statement also.
- Applications that depend on basic features of the blockchain can thus be developed without asking anybody for permission or paying anyone. 3 is correct.

Q.15 (2022)

Q. With reference to Web 3.0, consider the following statements:

1. Web 3.0 technology enables people to control their own data.
2. In Web 3.0 world, there can be blockchain-based social networks.
3. Web 3.0 is operated by users collectively rather than a corporation

Which of the following given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only

(d) 1, 2 and 3

Answer: D

- Web 3.0 is in an internet where users own their own data, not corporations
- Global digital transactions are secure, online exchanges of information and value are decentralized.
- Instead of being owned by centralized institutions, the community will own the network.
- This means the governance, rule sets, and general operation of the platform will move back to the users, as will the larger share of the potential profits.
- Blockchain is imperative to this, working as a digital ledger.
- Users will have their profile on their own Web 3.0 wallet which can be utilized across any decentralized platform.

Q.16 (2022)

Q. With reference to “Software as a Service”, Consider the following statements.

1. SaaS buyers can customise the user interface and can change data fields.
2. SaaS users can access their data through their mobile devices.
3. Outlook, Hotmail and Yahoo! Mail are forms of SaaS.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only

(d) 1, 2 and 3

Answer: D

- Software as a service (SaaS) is a software distribution model in which a cloud provider hosts applications and makes them available to end users over the internet.
- The public cloud provider manages all the hardware and traditional software, including middleware, application software, and security.
- SaaS makes it easy to "mobilise" the workforce because users can access SaaS apps and data from any Internet-connected computer or mobile device.
- Common examples are email, calendaring and office tools (such as Microsoft Office 365).
- Buyers can customize the user interface (UI) to change the look and feel of the program, as well as modify specific areas, such as data fields, to alter what data

Q.17 (2022)

Q. Which one of the following is the context in which the term “qubit” is mentioned?

(a) Cloud Services

(b) Quantum Computing

(c) Visible Light Communication Technologies

(d) Wireless Communication Technologies

Answer: B

- A qubit (or quantum bit) is the quantum mechanical analogue of a classical bit.
- In classical computing the information is encoded in bits, where each bit can have the value zero or one.
- In quantum computing the information is encoded in qubits.
- Qubits can be in a state of 0 or 1 or (unlike a classical bit) in a linear combination of both states.

Q.18 (2022)

Q. Consider the following communication technologies:

1. Closed-circuit Television
2. Radio Frequency Identification
3. Wireless Local Area Network

Which of the above are considered of Short-Range devices/technologies?

- (a) 1 and 2 only.
- (b) 2 and 3 only
- (c) 1 and 3 only

(d) 1, 2 and 3

Answer: D

- Closed Circuit Television or CCTV is a short range device.
- Radio-frequency identification (RFID) that is being used for Toll collection is a short range device.
- A Wireless Local Area Network (WLAN) is a wireless computer network that links two or more devices using wireless communication within a limited area such as a home, school, computer laboratory, campus, or office building.

Q.19 (2014)

Q. In addition to fingerprint scanning, which of the following can be used in the biometric identification of a person?

1. Iris scanning
2. Retinal scanning
3. Voice recognition

Select the correct answer using the code given below.

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only

d) 1, 2 and 3

Answer: D

- Biometric verification is any means by which a person can be uniquely identified by evaluating one or more distinguishing biological traits.
- Unique identifiers include fingerprints, hand geometry, earlobe geometry, retina and iris patterns, voice waves, DNA, and signatures.

Q.20 (2015)

Q. With reference to 'Near Field Communication (NFC) Technology', which of the following statements is/are correct?

1. It is a contactless communication technology that uses electromagnetic radio fields
2. NFC is designed for use by devices which can be at a distance of even a metre from each other
3. NFC can use encryption when sending sensitive information.

Select the correct answer using the code given below.

- | | |
|------------------------|--------------|
| a) 1 and 2 only | b) 3 only |
| c) 1 and 3 only | d) 1,2 and 3 |

Answer: C

- Near-field communication is a wireless technology that allows users to make payments by placing a compatible device like a smartphone or payment card within a few centimetres of another compatible device like a terminal, tablet or another smartphone.
- Near-field communication transmits data through electromagnetic radio fields and is the technology behind payment services like Google Wallet and Apple Pay.
- Both devices must contain NFC chips for an NFC transaction to take place.